

Self-Supervised Masking for Interpretable and Robust Facial Emotion Recognition



**A thesis submitted for the partial fulfillment of the requirements for
the Master in Data Science**

**Submitted to the
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By

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Bhim Lama

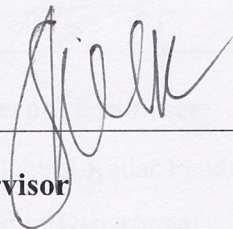
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This is to recommend that **Bhim Lama** has carried out the thesis entitled “**Self-Supervised Masking for Interpretable and Robust Facial Emotion Recognition**” for the partial fulfillment of the requirement of Master of Science (M.Sc.) Degree in Data Science under my supervision. To my knowledge, this work has not been submitted to any other institution for any academic degree. He has fulfilled all the requirements of the School of Mathematical Sciences, Institute of Science and Technology (IoST), Tribhuvan University, Nepal for the submission of the thesis for the Master of Science (M.Sc.) Degree in Data Science.



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Abstract

Facial emotion recognition (FER) is a critical component of human-computer interaction, with applications in mental health assessment, driver safety, and security systems. However, real-world FER systems face two persistent challenges: severe class imbalance that causes models to fail on rare emotions such as disgust and fear, and a lack of interpretability in black-box deep learning models. This thesis addresses both challenges by proposing **Mask-CAM** (Masked Class Activation Mapping), a lightweight and intrinsically interpretable convolutional neural network that learns a self-supervised soft mask to suppress background regions and focus on discriminative facial features without requiring external mask labels.

The proposed architecture extends a standard CAM network with a mask generator driven by competing losses: cross-entropy for emotion recognition, a sparsity penalty that encourages the mask to suppress as many regions as possible, and a smoothness (total variation) penalty that ensures contiguous, blob-like masks. A warm-up schedule gradually introduces the sparsity pressure during training. Experiments on FERPlus (in-distribution) and RAF-DB (Real-world Affective Faces Database, cross-dataset) demonstrate that Mask-CAM achieves competitive accuracy (78.85% on FERPlus, 67.50% on RAF-DB) while attaining the highest macro F1 on FERPlus (65.60%) and strong macro F1 on RAF-DB (46.71%). Crucially, Mask-CAM is the only model to detect disgust on RAF-DB ($F1 = 0.05$) and improves disgust F1 on FERPlus from 0.00 to 0.38. Overlay accuracy using facial landmarks confirms that the mask focuses on facial regions in 99.3% of FERPlus and 100% of RAF-DB images, and the mask density adapts to emotion difficulty—sparser for happy, denser for disgust.

With just 0.48 million parameters, which is six times fewer than a comparable black-box baseline, Mask-CAM is small enough to run in real time on resource-constrained devices. Test-time augmentation (TTA) further boosts accuracy to 80.05% (FERPlus) and 68.16% (RAF-DB), with the smallest cross-dataset drop among all compared models. McNemar’s test shows that Mask-CAM’s accuracy is not significantly different from the baselines ($p > 0.05$), confirming that interpretability is added without sacrificing

performance. This work demonstrates that a self-supervised soft mask can simultaneously provide built-in interpretability and improved recognition of under-represented emotions.

Keywords: *Facial Emotion Recognition, Self-supervised Learning, Interpretability, Soft Mask, Sparsity, Smoothness, CAM, FERPlus, RAF-DB*

शोधसार

अनुहारको भावना पहिचान (Facial Emotion Recognition) मानव-संगणक अन्तरक्रिया, मानसिक स्वास्थ्य मूल्याङ्कन, र सवारी सुरक्षाका लागि अत्यन्त महत्वपूर्ण क्षेत्र हो। तथापि, वास्तविक-विश्वका भावना पहिचान प्रणालीहरू दुई प्रमुख चुनौतीहरूले ग्रस्त छन्: गम्भीर वर्ग असंतुलन (class imbalance) जसले मोडेलहरूलाई दुर्लभ भावनाहरू जस्तै घृणा र डर पहिचान गर्न असफल बनाउँछ, र कालो-बाकस (black-box) गहिरो सिकाइ मोडेलहरूमा व्याख्यायोग्यताको अभाव। यस शोध प्रबन्धले दुवै चुनौतीलाई सम्बोधन गर्दै Mask-CAM (मास्कड क्लास एक्टिभेसन म्यापिङ) प्रस्ताव गर्दछ, जुन एक हल्का र आन्तरिक रूपमा व्याख्यायोग्य (intrinsically interpretable) कन्भोल्युसनल तंत्रिका सञ्जाल हो जसले पृष्ठभूमिलाई दबाउन र विशिष्ट अनुहार विशेषताहरूमा ध्यान केन्द्रित गर्न स्व-पर्यवेक्षित (self-supervised) सफ्ट मास्क सिक्छ, बाह्य मास्क लेबलहरूको आवश्यकता बिना नै।

प्रस्तावित आर्किटेक्चरले मानक CAM सञ्जाललाई मास्क जेनेरेटरको साथ विस्तार गर्दछ जसले प्रतिस्पर्धी हानिहरूद्वारा संचालित स्थानिय मास्क उत्पादन गर्दछ: भावना पहिचानको लागि क्रस-एन्ट्रोपी, मास्कले सकेसम्म धेरै क्षेत्रहरूलाई दबाउन प्रोत्साहित गर्ने स्पार्सिटी दण्ड, र निरन्तर, ब्लब-जस्तो मास्कहरू सुनिश्चित गर्ने स्मूथनेस (टोटल भेरिएसन) दण्ड। तालिमको क्रममा स्पार्सिटी दबावलाई क्रमशः परिचय गराउन वार्म-अप तालिका प्रयोग गरिन्छ। FERPlus (इन-डिस्ट्रिब्युसन) र RAF-DB (क्रस-डेटासेट) मा गरिएका प्रयोगहरूले Mask-CAM ले प्रतिस्पर्धी शुद्धता (FERPlus मा ७८.८५%, RAF-DB मा ६७.५०%) हासिल गरेको देखाउँछन्, साथै FERPlus मा उच्चतम macro F1 (६५.६०%) र RAF-DB मा बलियो macro F1 (४६.७९%) प्राप्त गरेको छ। विशेष गरी, यो RAF-DB मा घृणा पहिचान गर्ने एक मात्र मोडेल हो ($F1 = 0.05$) र FERPlus मा घृणाको F1 लाई ०.०० बाट ०.३८ मा सुधार गर्दछ। अनुहारका ल्यान्डमार्कहरूको प्रयोग गरी गरिएको ओभरले शुद्धता विश्लेषणले Mask-CAM ले ९९.३% FERPlus र १००% RAF-DB छविहरूमा अनुहार क्षेत्रमा केन्द्रित रहेको पुष्टि गर्दछ, र मास्कको घनत्व भावनाको कठिनाइ अनुसार अनुकूलन हुन्छ 'सजिलो भावनाका लागि पातलो, कठिन भावनाका लागि बाक्लो'।

केवल ०.४८ मिलियन प्यारामिटरहरू 'तुलनात्मक कालो-बाकस आधारभूत मोडेल भन्दा छ गुणा कम'संग, Mask-CAM स्रोत-सीमित यन्त्रहरूमा वास्तविक-समय प्रयोगको लागि उपयुक्त छ। परीक्षण-समय वर्धन (test-time augmentation) ले शुद्धतालाई FERPlus मा

८०.०५% र RAF-DB मा ६८.१६% सम्म बढाउँछ, र क्रस-डेटासेट ड्रप सबैभन्दा सानो रहन्छ। म्याक्नेमार परीक्षण (McNemar's test) ले Mask-CAM को शुद्धता आधारभूत मोडेलहरूसँग सांख्यिकीय रूपमा भिन्न नभएको देखाउँछ ($p > 0.05$), जसले व्याख्यायोग्यता प्रदर्शनको क्षति बिना नै थपिएको पुष्टि गर्दछ। यस कार्यले स्व-पर्यवेक्षित सफ्ट मास्कले मानक CAM सञ्जालमा कुनै अतिरिक्त गणनात्मक लागत बिना नै आन्तरिक व्याख्यायोग्यता र कम-प्रतिनिधित्व भावनाहरूको सुधारिएको पहिचान एकैसाथ प्रदान गर्न सक्छ भन्ने प्रमाणित गर्दछ, जसले वास्तविक-विश्व अनुप्रयोगहरूमा पारदर्शी र विश्वसनीय FER प्रणालीहरूको मार्ग प्रशस्त गर्दछ।

मुख्य शब्दहरू: *Facial Emotion Recognition, Self-Supervised Learning, Interpretability, Soft Mask, Sparsity, Smoothness, CAM, FERPlus, RAF-DB*

LIST OF ABBREVIATIONS AND ACRONYMS

ADAM	Adaptive Moment Estimation
ANN	Artificial Neural Network
AU	Action Unit
AUC	Area Under the Curve
BERT	Bidirectional Encoder Representations from Transformers
CAM	Class Activation Mapping
CNN	Convolutional Neural Network
CV	Cross-Validation
DNN	Deep Neural Network
FER	Facial Emotion Recognition
F1	F1-Score
Grad-CAM	Gradient-weighted Class Activation Mapping
HCI	Human-Computer Interaction
IoST	Institute of Science and Technology
L2	L2 Regularization
MAE	Mean Absolute Error
Mask-CAM	Masked Class Activation Mapping
MIM	Masked Image Modeling
ML	Machine Learning
MLP	Multilayer Perceptron
MSE	Mean Squared Error

LIST OF ABBREVIATIONS AND ACRONYMS

RAF-DB	Real-world Affective Faces Database
ReLU	Rectified Linear Unit
RMSE	Root Mean Square Error
ROC	Receiver Operating Characteristic
SD	Standard Deviation
SGD	Stochastic Gradient Descent
SHAP	SHapley Additive exPlanations
SSL	Self-Supervised Learning
TTA	Test-Time Augmentation
TU	Tribhuvan University
ViT	Vision Transformer
XAI	Explainable Artificial Intelligence

LIST OF SYMBOLS

$\mathcal{L}$	loss function
θ	model parameters
α	learning rate / regularization parameter
β	momentum parameter
γ	masking ratio
λ	regularization coefficient
ϵ	small constant for numerical stability
$\mathcal{D}$	dataset / training data
$\mathcal{M}$	masked image
$\mathbf{x}$	input image
$\mathbf{y}$	true label / ground truth
$\hat{\mathbf{y}}$	predicted label
$\mathbf{W}$	weight matrix
$\mathbf{b}$	bias term
σ	activation function (e.g., sigmoid)
∇	gradient operator
ℓ	loss value
p	probability / confidence score
k	number of classes
n	batch size / number of samples
d	feature dimension
$[MASK]$	mask token
L	number of layers
H	hidden dimension size
T	temperature parameter (in softmax)
A	attention matrix
K	kernel size
S	stride size
P	padding size
F	number of filters / feature maps

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Chapter 1

Introduction

1.1 Background

Facial expressions are one of the best ways to express emotional states and intentions as they are natural and universal signals (Li and Deng, 2022). They have many applications in entertainment, transportation, health, education and others (Ullah et al., 2024).

In Li and Deng (2022), Ekman and Friesen mention that there are six basic emotions: anger, disgust, fear, happiness, sadness, and surprise. However, contempt was later added as a seventh basic emotion. Advanced research in neuroscience and psychology has pointed out that these six emotions are not universal but rather culture-specific (Jack et al., 2012).

FER systems can be classified into two main types based on their feature representations. The first is static image FER, which encodes spatial information from a single image, and the second is dynamic sequence FER, which leverages the temporal relationship among consecutive frames of a facial expression sequence (Li and Deng, 2022). Most traditional methods relied on handcrafted features or shallow learning techniques such as Local Binary Patterns (LBP) (Li and Deng, 2022). However, recent competitions like the FER2013 challenge (Goodfellow et al., 2013) and the Emotion Recognition in the Wild (EmotiW) challenge (Dhall et al., 2012) have provided a large amount of training data from real-life scenarios, encouraging a shift from lab-controlled FER to in-the-wild FER.

There are many state-of-the-art models achieving great performance (Li and Deng, 2022; Ullah et al., 2024). However, deep learning for FER still faces challenges, such as the need for large amounts of training data (Li and Deng, 2022; Ullah et al., 2024) and high inter-subject variation due to factors like age, gender, and ethnicity, as well as variations in pose, lighting, and occlusion, which make it difficult for the model to learn discriminative features (Li and Deng, 2022).

One of the main problems in large-scale FER is class imbalance (Li and Deng, 2022). As mentioned in the literature, expressions like happiness and neutrality are common and they appear thousands of times in the datasets, whereas disgust and fear are the emotions that are harder to find (Li et al., 2017). This imbalance causes the models to focus on frequent classes and ignore the ones that occur less frequently. As a result, models with high accuracy sometimes fail to detect emotions like disgust (Li et al., 2017; Barsoum et al., 2016).

The other critical issue is the lack of interpretability in deep learning models. Selvaraju et al. (2017) explain that the “black box” nature of deep neural networks makes them hard to interpret. Interpretability is important to build trust in intelligent systems. Therefore, we need models that can explain why they make certain decisions (Ribeiro et al., 2016).

To address these challenges, various methods have been proposed. For example, Barsoum et al. (2016) used crowd-sourced label distributions to handle uncertainty, and Wang et al. (2020) proposed a Self-Cure Network to suppress uncertainties. However, many of these methods rely heavily on labeled data and sometimes fail to generalize across different datasets. Thus, the interpretability problem remains a major concern.

Self-supervised learning is a technique where the models teach themselves without any external labels. According to Gao and Patras (2023), self-supervised pretraining is effective in improving performance on various tasks such as image classification and object detection. Among the self-supervised learning approaches, masked image modeling (MIM) has been particularly successful. He et al. (2022) showed that Masked Autoencoders (MAE) can learn important visual representations by masking random patches of the input image and reconstructing the missing parts. Similarly, Alshami et al. (2025) proposed a method called AIM that uses self-supervised masking to improve interpretability by encouraging the models to focus on genuine features rather than spurious image features.

1.2 Statement of the Problem

There are two core challenges that need to be solved.

(i) Class Imbalance. Class imbalance in FER datasets causes models to focus on frequent emotions like happiness and to fail to recognize rare emotions such as disgust (Li et al., 2017).

(ii) Lack of Interpretability. Most of the models with high performance are black-boxes which provide no evidence of where they focused while making decisions. On the other hand, techniques like Grad-CAM and LIME give post-hoc explanations (Selvaraju et al., 2017; Ribeiro et al., 2016). They are computationally expensive and may not be trustworthy (Ribeiro et al., 2016; Alshami et al., 2025).

This research aims to address these problems by proposing Mask-CAM, a lightweight and intrinsically interpretable convolutional neural network. The model learns a self-supervised soft mask that suppresses unnecessary image regions, without requiring any external mask labels. Importantly, the goal of this work is not to surpass current state-of-the-art models in accuracy. Instead, we show that intrinsic interpretability can be added without sacrificing accuracy or incurring extra cost. We compare three architectures: a black-box CNN (BaselineCNN), a standard CAM-based model (CAMCNN), and our Mask-CAM, and evaluate their trade-offs on both in-distribution (FERPlus) and cross-dataset (RAF-DB) benchmarks.

1.3 Research Objectives

The primary objectives of this research are:

1. To develop Mask-CAM, a lightweight and interpretable model that learns to focus on the face without external mask labels.
2. To evaluate Mask-CAM against baselines on FERPlus and RAF-DB in terms of accuracy, interpretability, and efficiency.

1.4 Significance of the Study

This research is important both scientifically and practically. Scientifically, it provides a detailed comparison of three FER models: a black-box CNN, a standard CAM (Zhou

et al., 2016), and the proposed Mask-CAM, which adapts the self-supervised masking mechanism of AIM (Alshami et al., 2025). It demonstrates that a self-supervised soft mask can simultaneously improve interpretability and cross-dataset generalization.

Practically, Mask-CAM shows that emotion recognition can be both accurate and interpretable. With just 0.48 million parameters (six times fewer than the baseline), it delivers competitive accuracy while providing visual interpretability. It performs well on the cross-dataset RAF-DB benchmark, confirming its robustness to domain shift and demonstrating significant improvement on rare emotions.

1.5 Scope and Limitations

Scope:

- The research focuses on static image-based facial emotion recognition using the FERPlus and RAF-DB datasets.
- The proposed Mask-CAM architecture is designed for the seven basic emotion categories (angry, disgust, fear, happy, neutral, sad, surprise).
- The model is evaluated on an in-distribution dataset (FERPlus) and a cross-dataset benchmark (RAF-DB).
- Interpretability is analyzed through overlay accuracy, qualitative visualizations, and mask density analysis.

Limitations:

- The research does not consider video-based facial emotion recognition.
- The model is evaluated on two datasets only; further validation on additional datasets would strengthen the findings.
- The self-supervised mask is class-agnostic; it does not provide class-specific masks.

- The mask generator is designed for convolutional neural networks, which may not be directly applicable to transformer-based architectures.
- The small sample size for disgust in the test sets limits the statistical significance of improvements on this class.

Chapter 2

Literature Review

2.1 Facial Emotion Recognition: An Overview

Facial emotion recognition is an important part of human-computer interaction. According to Li and Deng (2022), facial expression is one of the best ways human beings can express their emotional states and intentions. The field has evolved significantly from early handcrafted feature-based methods to modern deep learning approaches.

Traditional FER systems generally follow a three-stage workflow: face acquisition, facial feature extraction, and facial expression classification (Li and Deng, 2022; Li et al., 2017). In the initial stage, an automated detector locates the face and uses specific facial features, such as the eyes and mouth, to align the image to a standard template (Ullah et al., 2024). Next, feature extraction can be divided into methods that use handcrafted descriptors like LBP, HOG, or Gabor features, and Action Unit (AU)-based methods that detect emotion through muscle movements (Li et al., 2017). Finally, the extracted features are processed by a classifier into specific emotional labels (Ullah et al., 2024).

Two major changes caused the shift from traditional methods to modern deep learning in facial expression recognition. First, large-scale contests such as FER2013 and EmotiW provided massive amounts of real-world image data, allowing systems to move beyond the limitations of lab-controlled images and learn from “in-the-wild” scenarios (Goodfellow et al., 2013; Li and Deng, 2022). Second, significant improvements in computing power, such as GPUs, and advanced neural network designs enabled researchers to apply deep learning effectively (Khairuddin and Chen, 2021). As a result, these modern systems have achieved much higher accuracy than traditional methods (Ullah et al., 2024).

Deep neural networks require a large amount of training data to avoid overfitting, but current databases are often insufficient for training such networks (Li and Deng, 2022; Zhang et al., 2018). Furthermore, factors like age, gender, and ethnic background cause inter-subject differences (Ahmad et al., 2022). Beyond identity bias, unconstrained FER

is further complicated by variations in head pose, ambient illumination, and physical occlusions (Ullah et al., 2024). For these reasons, deep networks must handle high intra-class variability to extract discriminative, expression-specific representations.

The fundamentals of FER, its architectural elements, applications, and future directions are discussed by Ullah et al. (2024). The authors identify two main challenges in in-the-wild conditions: pose variation and occlusion. They also highlight that novel deep learning models are needed to address cultural variations in emotional expressions (Jack et al., 2012).

Convolutional Neural Networks (CNNs) have emerged as the dominant architecture for Facial Expression Recognition (FER) because they can learn complex features directly from raw image pixels (Khairuddin and Chen, 2021; Ullah et al., 2024). A notable example is the work of Khairuddin and Chen (2021), who employed a VGGNet architecture to achieve state-of-the-art accuracy on the FER2013 dataset. They demonstrated that their model successfully identified and focused on the critical facial regions.

As networks grow deeper, they encounter problems such as vanishing gradients and increased computational cost. He et al. (2016) addressed these issues by introducing residual connections, which allow gradients to flow directly through the network and enable the training of very deep architectures. Their ResNet architecture has become a foundational building block for many FER models. Additionally, Hu et al. (2018) proposed Squeeze-and-Excitation (SE) networks, which introduce channel-wise attention to recalibrate feature responses. SE blocks have been shown to improve FER performance by enabling the network to focus on the most informative channels.

Attention mechanisms have also played an important role in FER (Ullah et al., 2024; Wang et al., 2020). Class Activation Mapping (CAM), proposed by Zhou et al. (2016), identifies discriminative image regions used by CNNs without requiring fully-connected layers. CAM demonstrated that global average pooling enables the network to localize objects despite being trained only on image-level labels. This work laid the foundation for interpretable FER models (Zhou et al., 2016).

In terms of available data, the Extended Cohn-Kanade (CK+) database is the most widely used collection of facial images, created in a controlled laboratory setting

for testing emotion-recognition systems (Li and Deng, 2022). It features 593 video clips from 123 different people, each showing one of seven basic emotions, including happiness, sadness, anger, and contempt (Lucey et al., 2010). The MMI database is another laboratory-controlled collection, but it presents difficult challenges (Li and Deng, 2022). It contains 326 video sequences from 32 individuals, with over 200 of those sequences labeled with six basic emotions (Valstar and Pantić, 2010). This database is particularly tough because people express the same emotion in different ways that can distract the model. The Japanese Female Facial Expression (JAFPE) database consists of 213 images of 10 Japanese women who were asked to pose with specific expressions (Li and Deng, 2022). For each of the six basic emotions, every woman provided three or four different photos (Lyons et al., 1998). Although it is smaller than other collections, JAFPE is a significant resource for studying facial cues within a specific cultural and gender group (Lukac et al., 2023).

The FER2013 dataset was introduced during the ICML 2013 Challenges in Representation Learning. According to Goodfellow et al. (2013), FER2013 is a large-scale database collected automatically by the Google image search API. It contains 28,709 training images, 3,589 validation images, and 3,589 test images with seven expression labels. The images were processed using OpenCV for face recognition, and human labelers rejected the incorrectly labeled ones. Approved images were resized to 48×48 pixels and converted to grayscale. The dataset contains 35,887 images. The authors discovered that even humans struggle with the FER2013 dataset, correctly identifying expressions only about 65% (plus or minus 5%) of the time. This shows that the task is quite difficult for people as well.

FERPlus is the extension of FER2013 where each image was re-labeled by 10 crowd taggers (Barsoum et al., 2016). The authors demonstrated that it is better to use the complete variety of opinions provided by all taggers instead of just picking the popular emotion label. RAF-DB (Real-world Affective Face Database) contains around 30,000 facial images, and each image was labeled about 40 times to filter out unreliable labels (Li et al., 2017). It is the first database containing compound expressions. By comparing CK+ and RAF-DB, the authors found that facial movements in everyday life are much more varied, and often differ from lab-controlled expressions.

Class imbalance is a fundamental challenge in facial emotion recognition, particularly for in-the-wild datasets. Expressions like happiness and neutrality appear thousands of times, while important emotions like disgust and fear are extremely rare (Li et al., 2017). This leads to a model focusing on frequent classes and ignoring the rare ones. Li and Deng (2022) explain that because some facial expressions are much harder to collect and label than others, AI models often develop a significant bias. They point out that while finding and tagging a smile is simple, capturing authentic “disgust” is a major challenge. Consequently, when these models are trained using standard rules, they naturally prioritize the most common emotions. This often results in a system that appears highly accurate overall but almost completely fails to recognize rare emotions. This problem is clearly seen in the RAF-DB dataset, where a standard VGG model was found to be over 85% accurate at identifying happiness but reached only 27.50% accuracy for disgust (Li et al., 2017).

Several approaches have been proposed to address class imbalance in FER. Wang et al. (2020) proposed a Self-Cure Network that suppresses uncertainties. The SCN consists of three modules: self-attention importance weighting, ranking regularization, and relabeling. A self-attention importance weighting module gives a score to highlight clear examples and ignore the confusing ones. Next, a ranking regularization module sorts these scores, and finally a relabeling module identifies mislabeled samples and corrects their labels. The authors validated their method on RAF-DB, AffectNet, and FERPlus, achieving state-of-the-art results. Barsoum et al. (2016) demonstrated that using the full label distribution from crowd-sourced annotations improved FER performance. After comparing four different strategies—majority voting, multi-label learning, probabilistic label drawing, and cross-entropy loss—they found that the final two approaches were the most effective for training the model. Additionally, they found that using 10 different taggers per image made the data more reliable. Lin et al. (2017) proposed focal loss for dense object detection, which addresses class imbalance by adding a modulating factor to the cross-entropy loss that down-weights easy examples. By doing so, the model is forced to focus on a small set of hard examples. This approach is very useful for FER class imbalance.

The “black box” nature of deep neural networks has led to extensive research on interpretability. Selvaraju et al. (2017) note that interpretability matters because in order

to build trust in intelligent systems, we must build transparent models that can explain their predictions. Interpretability is useful at three different stages of AI evolution: first, it helps researchers identify and fix errors when systems are still weak; second, it builds user trust once AI becomes reliable; and finally, it allows superior AI to teach humans how to make better choices.

Grad-CAM (Gradient-weighted Class Activation Mapping) was introduced by Selvaraju et al. (2017) to generate visual explanations for many types of CNN models. It uses mathematical gradients to create heatmaps that highlight the most important regions of the image. By combining this with other visualization techniques, the authors created Guided Grad-CAM, which provides highly detailed, class-specific images to help users understand exactly why a model made a certain decision. LIME (Local Interpretable Model-agnostic Explanations) was developed by Ribeiro et al. (2016). It is a flexible tool that explains a model's decision by creating a simple and understandable model around a single prediction. The authors also introduced SP-LIME, which picks out the most representative examples to help users decide if they can trust the model overall. Tests showed that even non-experts could use LIME to identify better-performing systems.

Alshami et al. (2025) introduced AIM (Amending Inherent Interpretability via Self-Supervised Masking), a method that addresses the trade-off between interpretability and accuracy in deep neural networks. AIM encourages the model to focus on genuine parts of an image instead of misleading ones, without additional annotations. The main idea is that when the model is forced to select only a limited set of image regions, it will naturally choose the most important features. AIM uses a top-down approach across multiple levels to filter out irrelevant features. It uses a mask estimator that creates a simplified map to highlight important regions. The researchers employed the Gumbel-Softmax trick, which allows the model to learn to suppress or reveal different facial regions during training. Finally, the model uses a sparsity loss that encourages the network to ignore as many regions as possible while still maintaining high accuracy. Researchers tested AIM across different types of image datasets including ImageNet100, HardImageNet, ImageWoof, Waterbirds, TravelingBirds, and CUB-200. Their results were accurate and easier for humans to understand. The Energy Pointing Game (EPG) score was used to demonstrate that the model correctly focused on the main object.

AIM is highly effective for emotion recognition because models often get confused by irrelevant features, rather than looking at important facial regions. The proposed Mask-CAM model builds on the AIM framework by adding a Class Activation Mapping (CAM) tool (Zhou et al., 2016), providing two levels of interpretability: a class-agnostic soft mask that shows suppressed regions, and a CAM revealing which regions contributed to the decision.

Khairuddin and Chen (2021) utilized saliency maps to visualize how deep learning models process facial images. Their method successfully identified and visualized the important facial regions for each emotion.

Lukac et al. (2023) explored how cultural and regional biases can confuse FER models. They proposed a Multi-Cues Emotion Model (MCAM) that integrates image features, action units, micro-expressions, and macro-expressions. They found that identifying emotions in different regions relies on unique and group-specific traits. Their work underscores the importance of diverse training data and cross-cultural validation for building fair and reliable facial recognition systems.

He et al. (2022) introduced Masked Autoencoders (MAE), a self-supervised method that learns visual representations by reconstructing masked image patches. The system works by hiding a large portion of an image (about 75%) and forcing the model to rebuild the missing parts pixel by pixel. To make this process highly efficient, they created a two-part asymmetric system: a main “encoder” that only studies the small visible pieces and a lightweight “decoder” that puts the whole picture back together. This unique design makes training three times faster and significantly more accurate than previous methods. Their approach set a new record, achieving 87.8% accuracy on the ImageNet-1K benchmark.

Gao and Patras (2023) proposed Facial Region Awareness (FRA), a method that helps the model to recognize faces by focusing on consistent global and local facial details. FRA ensures that a model recognizes specific regions like eyes, nose, and mouth consistently. Their method uses a Transformer decoder and “facial queries” to search the image and find these specific regions. FRA treats the task as a clustering problem so that it learns those parts without needing human labels. The tests show that FRA outperforms older models on many facial recognition tasks.

2.7 Research Gaps

Based on the literature review, several critical research gaps remain unaddressed in facial emotion recognition.

The major gap is the difficulty in maintaining a balance between interpretability and accuracy (Selvaraju et al., 2017; Alshami et al., 2025). Tools like Grad-CAM and LIME attempt to provide explanations after the model’s decision, but these “after-the-fact” methods may not always faithfully reflect the model’s actual reasoning (Ribeiro et al., 2016; Alshami et al., 2025). In contrast, models like CAM offer a transparent design but often sacrifice accuracy (Zhou et al., 2016; Selvaraju et al., 2017). Therefore, there is a strong need for FER systems that can provide built-in explanations without sacrificing performance (Alshami et al., 2025).

Class imbalance further complicates emotion recognition, as datasets often contain many examples of common expressions. Rare emotions like disgust remain much harder to identify than common expressions such as happiness or neutrality, leading to misleading performance evaluations. A model with very high overall accuracy may completely fail to recognize rare emotions. Hence, there is a pressing need for new methods that can improve performance on under-represented emotions (Li et al., 2017).

Self-supervised masking has proven effective for general computer vision tasks, yet it has not been widely applied to facial emotion recognition. Moreover, little research has focused on creating models whose interpretability is easily understood by humans. This represents a significant gap, as these techniques could enable models to learn reliable and clear features without requiring manual human labels (Gao and Patras, 2023).

Finally, cross-dataset generalization remains an ongoing challenge in FER. Models trained on a single dataset often fail to generalize to others, limiting their deployment in real-world scenarios where the environment and facial appearances vary considerably (Li and Deng, 2022).

This research introduces Mask-CAM, an efficient model designed to be intrinsically interpretable. It learns a soft mask that automatically suppresses irrelevant image regions, and it maintains competitive accuracy on both in-distribution and cross-dataset benchmarks while providing built-in interpretability.

Chapter 3

Research Methodology

This chapter presents the methodology used in this research. The workflow is organised into five important steps.

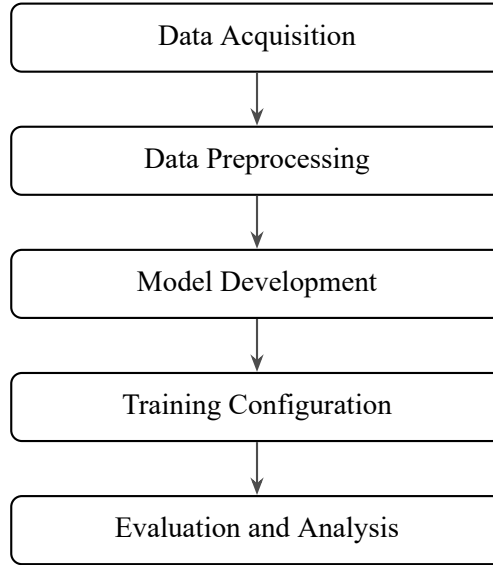


Figure 1: Methodology Flow chart.

3.1 Datasets

This research uses two publicly available datasets: FERPlus for training and in-distribution testing, and RAF-DB for cross-dataset evaluation.

3.1.1 FERPlus Dataset

FERPlus is an extension of the FER2013 dataset, where each image was re-labeled by 10 crowd taggers (Barsoum et al., 2016). The dataset contains 28,709 training images, 3,589 validation images, and 3,589 test images, all resized to 48×48 pixels and converted to grayscale.

Table 1: Distribution of emotions in the FERPlus training set.

Emotion	Number of Images	Percentage
Angry	3,995	13.9%
Disgust	436	1.5%
Fear	4,097	14.3%
Happy	7,215	25.1%
Sad	4,830	16.8%
Surprise	3,171	11.0%
Neutral	4,965	17.3%
Total	28,709	100.0%

The original FER2013 labels were provided with seven emotion categories in alphabetical order: angry, disgust, fear, happy, neutral, sad and surprise (Goodfellow et al., 2013). However, Barsoum et al. (2016) note that contempt is included in FERPlus, leading to 8 classes. For this research, the contempt class is excluded to maintain consistency with the seven basic emotions.

Label Mapping: The FERPlus CSV file uses the following label order: [neutral, happiness, surprise, sadness, anger, disgust, fear, contempt]. To align with the standard FER2013 alphabetical order [angry, disgust, fear, happy, neutral, sad, surprise], the following mapping is applied:

Table 2: Label mapping from FERPlus to standard FER2013 order.

Original FERPlus Index	Emotion	Mapped Index
0	neutral	4
1	happiness	3
2	surprise	6
3	sadness	5
4	anger	0
5	disgust	1
6	fear	2

Figure 2 shows sample images from the FERPlus dataset.



Figure 2: Sample images from FERPlus dataset.

3.1.2 RAF-DB Dataset

RAF-DB contains about 30,000 facial images from thousands of individuals, divided into a basic expression set and a compound expression set. This research uses only the basic expression subset, which provides 12,271 training images and 3,068 test images across seven emotion categories. Each image was labeled approximately 40 times and filtered using an EM algorithm to ensure label reliability (Li et al., 2017). RAF-DB is used exclusively as a cross-dataset benchmark to evaluate robustness under domain shift. The folder structure is organised by emotion categories.

3.1.3 Data Preprocessing

For both datasets, the following preprocessing steps are applied: images are resized to 48×48 pixels, pixel values are normalized to the range $[0, 1]$, and data augmentation is applied during training. The augmentation includes horizontal flipping, random rotation, affine transformations, color jitter, and grid occlusion (Khairuddin and Chen, 2021).

3.2 Model Architectures

Three models are implemented and compared in this research: BaselineCNN, CAM-CNN, and Mask-CAM. All three models share the same ImprovedBackbone for feature extraction.

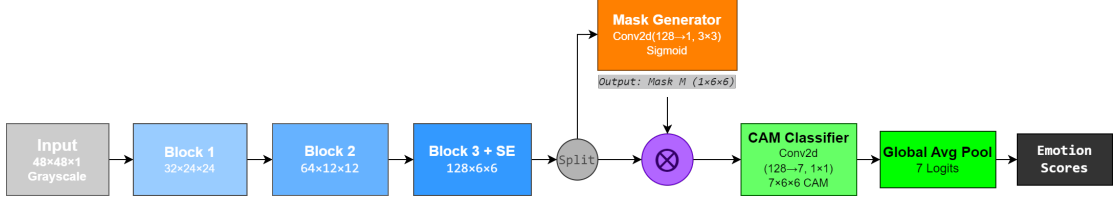


Figure 3: Architecture of Mask-CAM.

3.2.1 ImprovedBackbone

The ImprovedBackbone is designed to extract strong facial features. It consists of multiple convolutional layers with batch normalization to stabilize optimization and accelerate training (Ioffe and Szegedy, 2015), ReLU activation to introduce non-linear mapping capabilities (Nair and Hinton, 2010). It also has residual connections for easy gradient flow and smooth training of deeper representations (He et al., 2016). Additionally, Squeeze-and-Excitation (SE) blocks are added to recalibrate channel-wise feature responses (Hu et al., 2018). Finally, the architecture uses Dropout2d for spatial regularization to mitigate overfitting (Srivastava et al., 2014).

Table 3: Layer-by-layer architecture of ImprovedBackbone.

Layer	Input Shape	Output Shape	Parameters
Block1 (Conv + BN + ReLU)	$1 \times 48 \times 48$	$32 \times 24 \times 24$	3,200
Block2 (Conv + ResBlock)	$32 \times 24 \times 24$	$64 \times 12 \times 12$	55,552
Block3 (Conv + ResBlock + SE)	$64 \times 12 \times 12$	$128 \times 6 \times 6$	221,696
Total	—	—	280,448

3.2.2 BaselineCNN

The BaselineCNN is a standard black-box model with 2.84 million parameters. It uses the ImprovedBackbone to extract facial features, then the features are flattened and passed through two fully connected layers (with 512 and 7 units, respectively) to make the final emotion prediction.

3.2.3 CAMCNN

The CAMCNN is a lightweight model based on the classic architecture by Zhou et al. (2016). It replaces the fully connected layers with a 1×1 convolutional layer and global average pooling (GAP). This model has only 0.47 million parameters and generates a visual heatmap to show where it looked in an image.

3.2.4 Mask-CAM (Proposed)

Mask-CAM is the extension of CAMCNN which has a self-supervised mask generator that learns to suppress unwanted regions. The mask is applied to the feature map before the 1×1 convolutional layer, which forces the model to focus on important regions.

The mask generator consists of a 3×3 convolutional layer and a sigmoid activation function. The convolutional layer processes the 6×6 feature map and the sigmoid activation function helps to produce mask values in the range $[0, 1]$. The mask is applied element-wise to the feature maps:

$$\mathbf{F}_{\text{masked}} = \mathbf{F} \odot \mathbf{M}$$

where $\mathbf{F}$ is the feature map from the backbone, $\mathbf{M}$ is the soft mask (values in $[0, 1]$), and $\odot$ denotes element-wise multiplication with broadcasting across channels.

Table 4: Comparison of the three architectures.

Model	Description	Parameters
BaselineCNN	Black-box	2.84 M
CAMCNN	Standard CAM	0.47 M
Mask-CAM (proposed)	CAMCNN + self-supervised mask generator	0.48 M

3.3 Training Configuration

3.3.1 Loss Function

Mask-CAM uses a composite loss function designed to balance accuracy with mask interpretability. The total loss is defined as:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda \cdot \overline{\mathbf{m}} + \beta \cdot (|\Delta_x \mathbf{m}| + |\Delta_y \mathbf{m}|) \quad (1)$$

where:

- $\mathcal{L}_{\text{CE}}$ is the cross-entropy loss with label smoothing of 0.1 (Szegedy et al., 2016).
- $\lambda \cdot \overline{\mathbf{m}}$ is the sparsity loss, which penalizes the mean mask value, forcing the generator to identify the most sparse, discriminative regions (Alshami et al., 2025).
- $\beta \cdot (|\Delta_x \mathbf{m}| + |\Delta_y \mathbf{m}|)$ is the smoothness loss (total variation), which penalizes spatial variation to encourage contiguous, “blob-like” masks (Rudin et al., 1992).

The mean mask value $\overline{\mathbf{m}}$ is computed as:

$$\overline{\mathbf{m}} = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W m_{i,j} \quad (2)$$

The smoothness terms are computed as:

$$|\Delta_x \mathbf{m}| = \frac{1}{(H-1)W} \sum_{i=1}^{H-1} \sum_{j=1}^W |m_{i+1,j} - m_{i,j}| \quad (3)$$

$$|\Delta_y \mathbf{m}| = \frac{1}{H(W-1)} \sum_{i=1}^H \sum_{j=1}^{W-1} |m_{i,j+1} - m_{i,j}| \quad (4)$$

These three losses compete with each other: the classification loss encourages the mask to keep all information necessary for correct emotion recognition, the sparsity loss pushes the mask to suppress as many regions as possible, and the smoothness loss ensures that the kept regions form a contiguous, interpretable blob.

3.3.2 Hyperparameters

Table 5: Hyperparameter configuration.

Hyperparameter	Value
λ (sparsity)	0.0012
β (smoothness)	0.0005
λ warm-up	$0 \rightarrow 0.0012$ over first 10 epochs
Optimizer	Adam (Kingma and Ba, 2015), lr = 0.001
Scheduler	ReduceLROnPlateau (patience = 3, factor = 0.5)
Batch size	128
Epochs	50
Label smoothing	0.1
Dropout rate	0.3
Weight decay	0.0001

The model is optimized using Adam because it combines the benefits of adaptive gradient magnitudes and moving averages of gradients. A ReduceLROnPlateau scheduler is employed to reduce the learning rate when the validation loss plateaus (Khairuddin and Chen, 2021).

3.3.3 Data Augmentation

Training data is augmented using horizontal flipping, random rotation, affine transformations, and grid occlusion (Khairuddin and Chen, 2021). MixUp (Zhang et al., 2018) is applied to the baseline and CAMCNN but omitted for Mask-CAM to keep the mask generator stable.

3.3.4 Training Procedure

The mask generator and the backbone are trained simultaneously. A λ warm-up (threshold annealing) strategy is used to prevent the mask from collapsing before the classifier learns stable facial features. This method is similar to the AIM framework (Alshami et al., 2025).

$$\lambda_{\text{eff}}(t) = \begin{cases} \lambda \cdot \frac{t}{T_{\text{warmup}}}, & t \leq T_{\text{warmup}} \\ \lambda, & t > T_{\text{warmup}} \end{cases} \quad (5)$$

where $T_{\text{warmup}} = 10$ epochs.

3.3.5 Test-Time Augmentation (TTA)

Final evaluation uses horizontal-flip TTA. It averages the predictions from the original and flipped test images to improve robustness (Barsoum et al., 2016; Khairuddin and Chen, 2021):

$$\mathbf{p}_{\text{TTA}} = \frac{1}{2}(\mathbf{p}(\mathbf{x}) + \mathbf{p}(\mathbf{x}_{\text{flip}})) \quad (6)$$

where $\mathbf{p}(\mathbf{x})$ is the softmax probability vector for the original image and $\mathbf{p}(\mathbf{x}_{\text{flip}})$ is the probability vector for the horizontally flipped image.

3.4 Evaluation Metrics

3.4.1 Classification Metrics

The classification performance of all models is evaluated using the following metrics:

- **Accuracy:** Proportion of correct predictions across the test set.
- **Macro F1-Score:** Unweighted mean of per-class F1-scores, which gives equal weight to rare classes (e.g., disgust, fear).
- **Per-class F1-Score:** F1-score for each individual emotion category, used to identify specific strengths and weaknesses.
- **Confusion Matrix:** Visualises classification performance across all classes. The mean diagonal value is used as the primary metric for imbalanced datasets to prevent frequency bias (Li et al., 2017).

3.4.2 Interpretability Metrics

For Mask-CAM, interpretability is assessed through the following metrics:

- **Overlay Accuracy (EPG Score):** Similar to Alshami et al. (2025), the Energy Pointing Game (EPG) score is used to quantify spatial localization. It measures the proportion of mask attribution energy that falls within the ground-truth facial region.
- **Mask Density:** Mean mask value per emotion. Lower mask density indicates stronger suppression of irrelevant regions, demonstrating the model’s ability to rely on a sparse set of discriminative features.
- **Qualitative Visualization:** Mask overlays are visualised by superimposing the generated mask onto the input image. Suppressed regions are highlighted in red (mask ≈ 0) while kept regions remain in grayscale (mask ≈ 1), similar to the AIM framework.

3.4.3 Statistical Evaluation

We use a statistical method called McNemar’s test (McNemar, 1947) to evaluate whether Mask-CAM is truly better than BaselineCNN and CAMCNN.

Instead of just comparing overall accuracy, this test specifically focuses on how and where the models make mistakes on the same set of images. It does this by sorting the test results into a simple 2×2 grid showing four situations:

- When both models are correct
- When both models are incorrect
- When only the first model is correct
- When only the second model is correct

It follows a chi-squared distribution with one degree of freedom under the null hypothesis that the two models have equal error rates. A significant result ($p < 0.05$) indicates

a reliable difference in accuracy. To ensure the results are robust, we repeat every experiment five times using different random setups (seeds) and apply McNemar’s test to the combined predictions from all five runs.

3.4.4 Efficiency Metrics

- **Number of Parameters:** Total number of trainable parameters, used to evaluate the architectural overhead added by the mask generator.

3.5 Implementation Details

3.5.1 Software and Hardware

All experiments are implemented using PyTorch (version 1.12.0) as the deep learning framework. To make the learning process much faster, we trained and evaluated our models on Google Colab using a powerful NVIDIA Tesla T4 GPU with CUDA acceleration.

Table 6: Key libraries used in this research.

Library	Purpose
PyTorch	Deep learning framework
NumPy	Numerical computing
Matplotlib & Seaborn	Visualization (mask overlays, confusion matrices, training curves)
face_recognition	Facial landmark detection for overlay accuracy
scikit-learn	Metrics computation (F1-score, confusion matrix)
PIL / OpenCV	Image preprocessing and augmentation

3.5.2 Training Details

- **Random Seed:** A fixed random seed (e.g., 42) is used for all experiments to ensure reproducibility.
- **Validation Split:** 20% of the training data is reserved for validation, which is used for monitoring and hyperparameter tuning.

- **Early Stopping:** Not employed. All models are trained for exactly 50 epochs.
- **Dataset Storage:** Datasets are uploaded to Google Drive and mounted to the Colab runtime environment. The data loading pipeline reads images directly from the mounted drive.
- **Checkpointing:** Model checkpoints are saved periodically during training. This allows us to resume training from our last saved point.

3.5.3 Code Organization

The code is organised into separate blocks. This makes it easier to debug, modify, and understand.

Table 7: Code organization.

Component	Description
Data Loading	Custom DataLoader classes for FERPlus and RAF-DB, handling label mapping, preprocessing, and augmentation
Model Definitions	Separate classes for BaselineCNN, CAMCNN, and Mask-CAM
Training Loop	Standard training loop with learning rate scheduling, loss computation, and metric logging
Evaluation	Comprehensive evaluation script computing all classification and interpretability metrics
Visualization	Scripts for generating mask overlays, confusion matrices, and training curves

Chapter 4

Results and Discussion

4.1 Classification Performance

4.1.1 FERPlus (In-Distribution)

Table 8 presents the classification performance of all three models on the FERPlus test set.

Table 8: Classification performance on FERPlus.

Model	Accuracy	Macro F1
BaselineCNN	79.41%	60.08%
CAMCNN	79.16%	63.87%
Mask-CAM	78.85%	65.60%

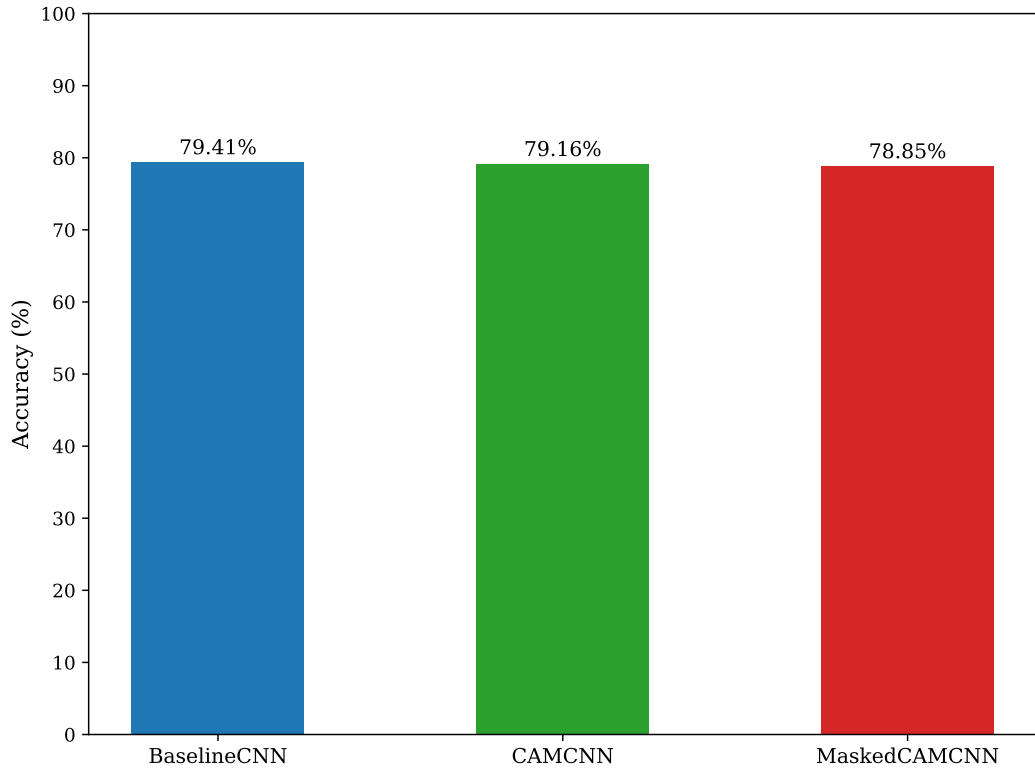


Figure 4: Overall accuracy on the FERPlus test set.

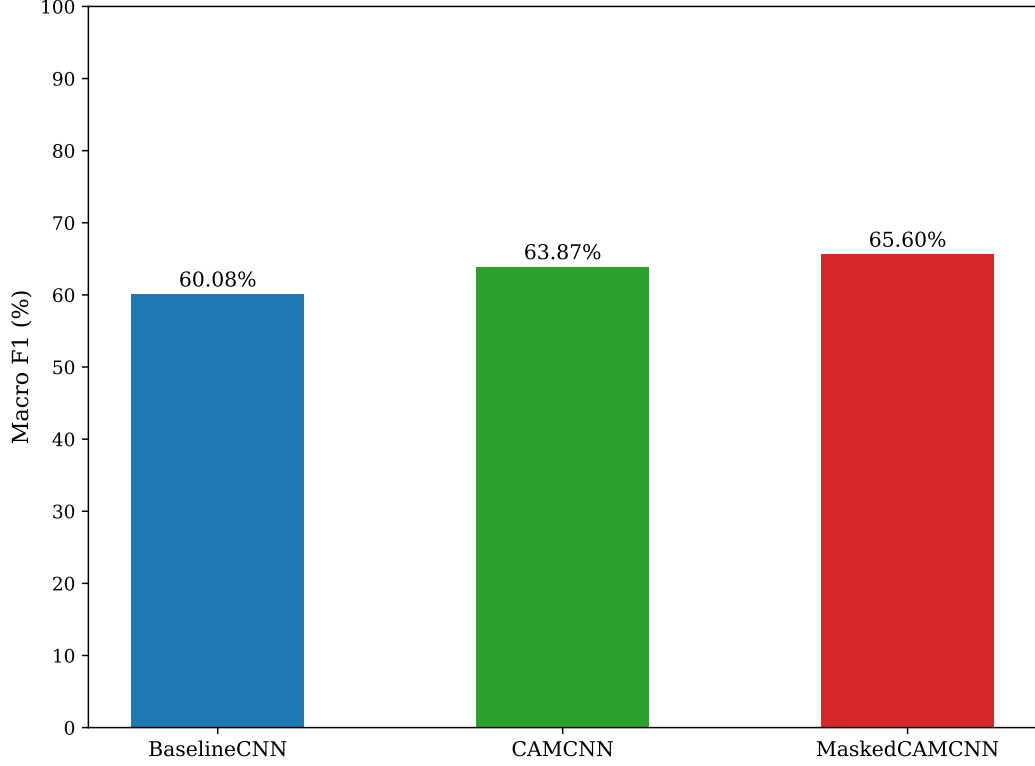


Figure 5: Macro F1 score on the FERPlus test set.

Mask-CAM achieves 78.85% accuracy, within 0.6% of the black-box baseline. More importantly, Mask-CAM achieves the highest macro F1-score (65.60%), indicating better performance on underrepresented classes.

Table 9: Per-class F1 scores on FERPlus.

Emotion	BaselineCNN	CAMCNN	Mask-CAM
Angry	0.73	0.70	0.71
Disgust	0.00	0.24	0.38
Fear	0.37	0.41	0.42
Happy	0.90	0.89	0.90
Sad	0.56	0.59	0.57
Surprise	0.82	0.80	0.80
Neutral	0.82	0.83	0.83

The most significant improvement is observed for disgust, where Mask-CAM achieves an F1-score of 0.38 compared to 0.00 for the baseline. Fear also improves from 0.37

(BaselineCNN) to 0.42 (Mask-CAM).

Confusion Matrices

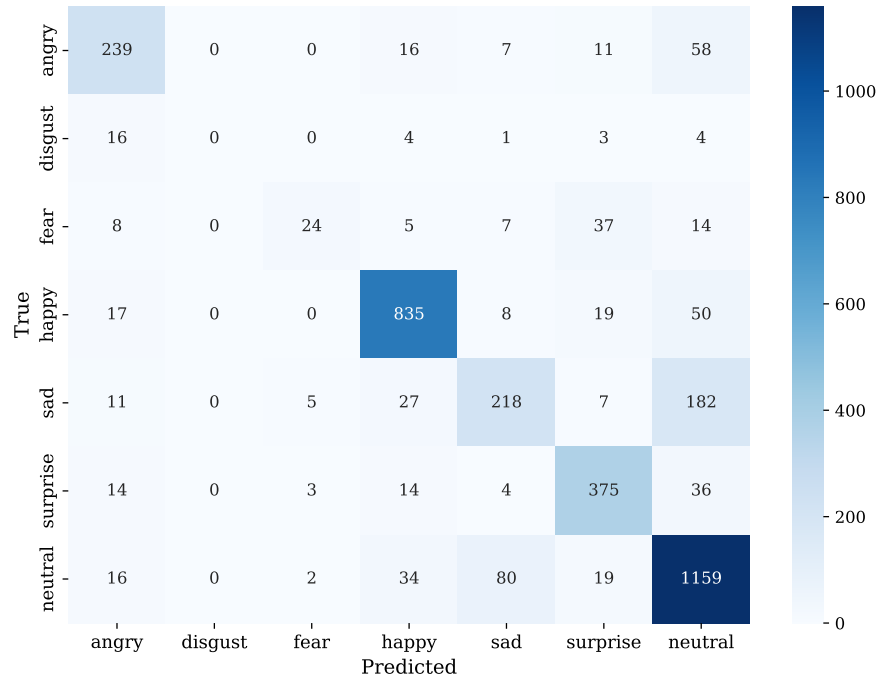


Figure 6: BaselineCNN on FERPlus (Acc: 79.41%).

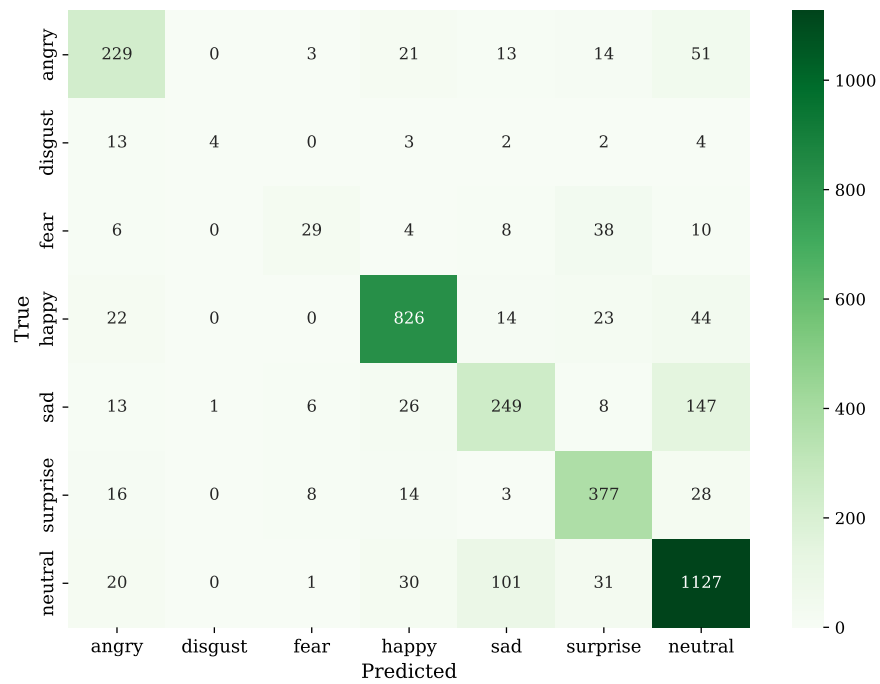


Figure 7: CAMCNN on FERPlus (Acc: 79.16%).

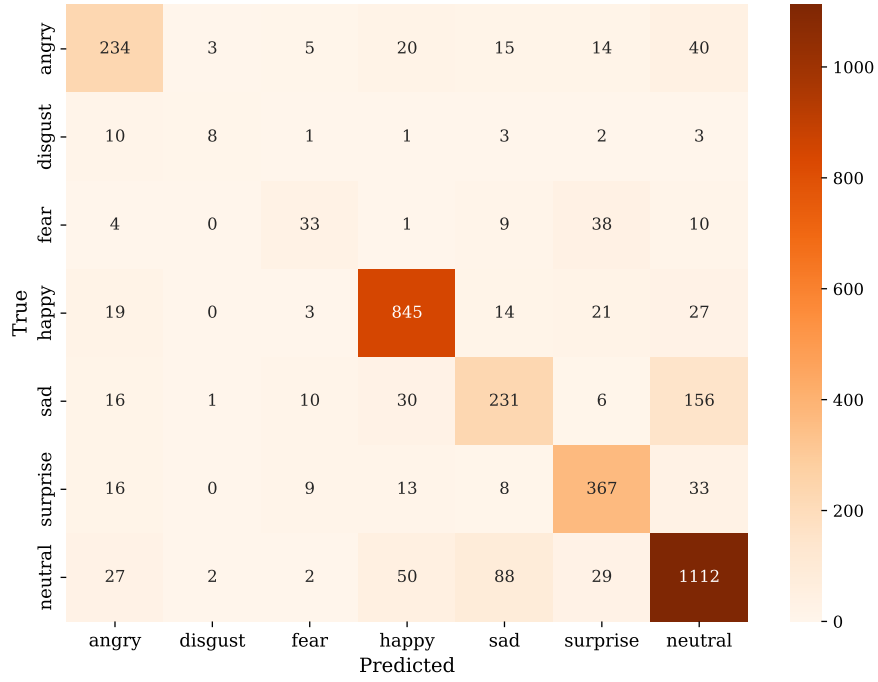


Figure 8: Mask-CAM on FERPlus (Acc: 78.85%).

On FERPlus, all models perform well on happiness and surprise. Mask-CAM shows reduced confusion for disgust, correctly classifying more samples compared to the baseline and CAMCNN.

4.1.2 RAF-DB (Cross-Dataset)

Table 10 presents the classification performance on the RAF-DB test set.

Table 10: Classification performance on RAF-DB.

Model	Accuracy	Macro F1
BaselineCNN	66.43%	43.29%
CAMCNN	67.28%	46.73%
Mask-CAM	67.50%	46.71%

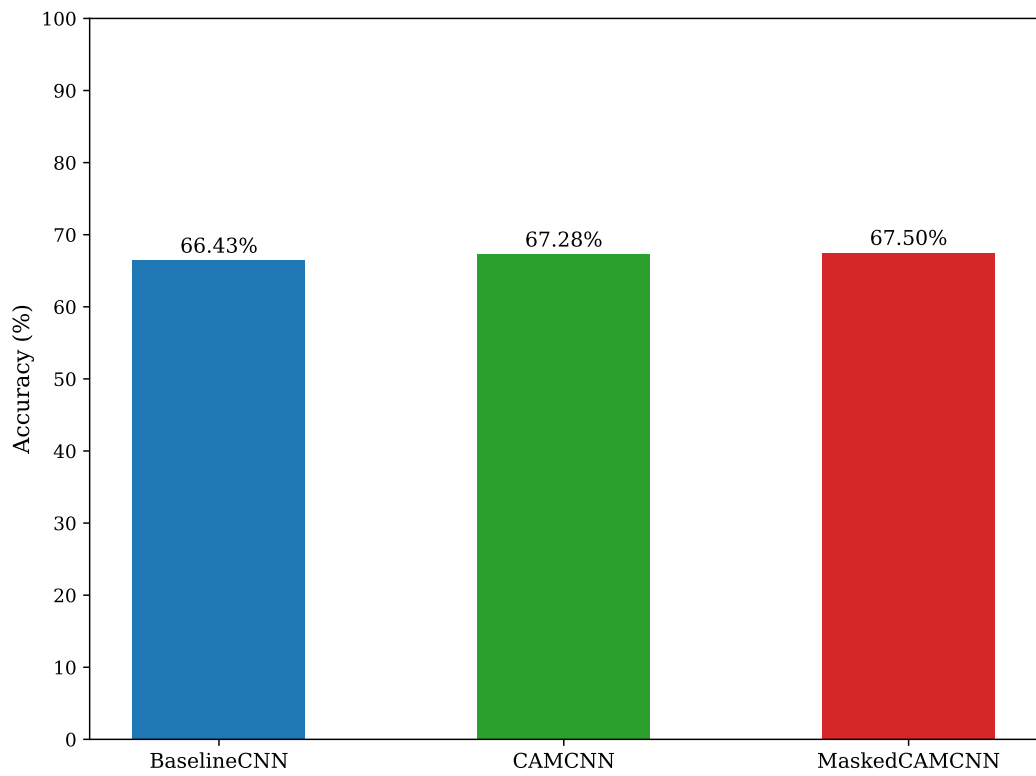


Figure 9: Overall accuracy on the RAF-DB test set.

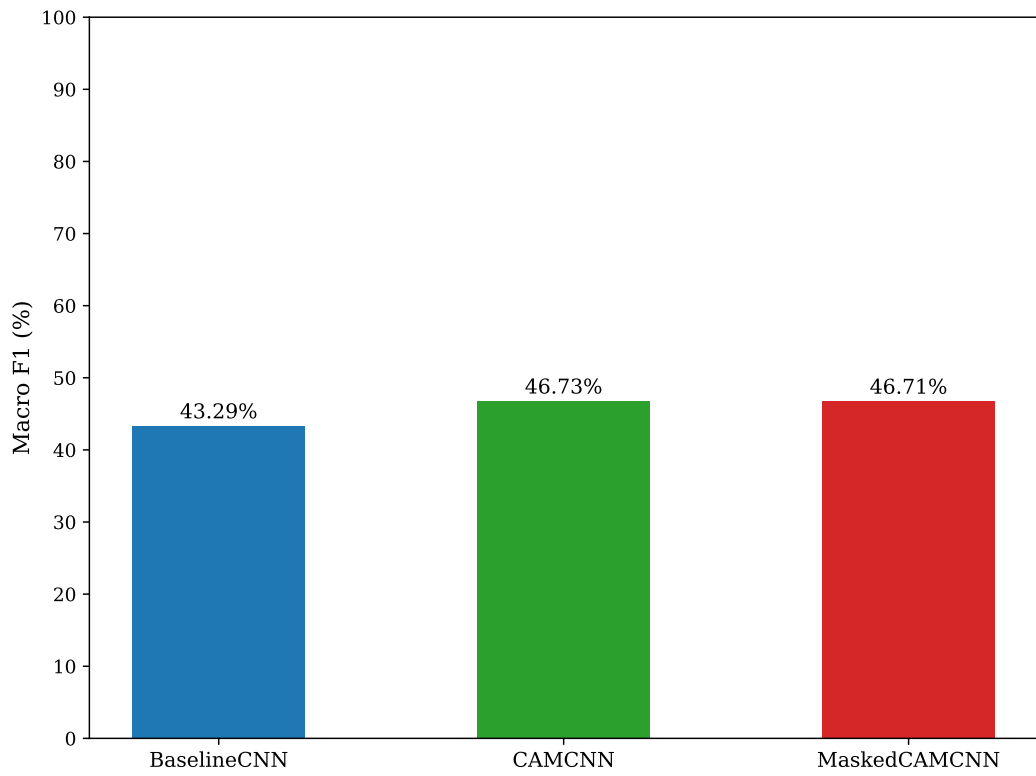


Figure 10: Macro F1 score on the RAF-DB test set.

Mask-CAM achieves the highest accuracy (67.50%) on RAF-DB, outperforming both the baseline and CAMCNN. This demonstrates the robustness of Mask-CAM under domain shift.

Table 11: Per-class F1 scores on RAF-DB.

Emotion	BaselineCNN	CAMCNN	Mask-CAM
Angry	0.55	0.56	0.57
Disgust	0.00	0.00	0.05
Fear	0.08	0.21	0.19
Happy	0.85	0.85	0.84
Sad	0.56	0.59	0.58
Surprise	0.36	0.43	0.40
Neutral	0.64	0.64	0.66

Mask-CAM is the only model that detects any disgust on RAF-DB ($F1 = 0.05$). Fear performance is substantially higher for CAMCNN (0.21) and Mask-CAM (0.19) compared to the baseline (0.08). This cross-dataset validation confirms that the improvements on rare emotions are not dataset-specific and that the self-supervised mask helps the model generalise genuine facial features.

Confusion Matrices

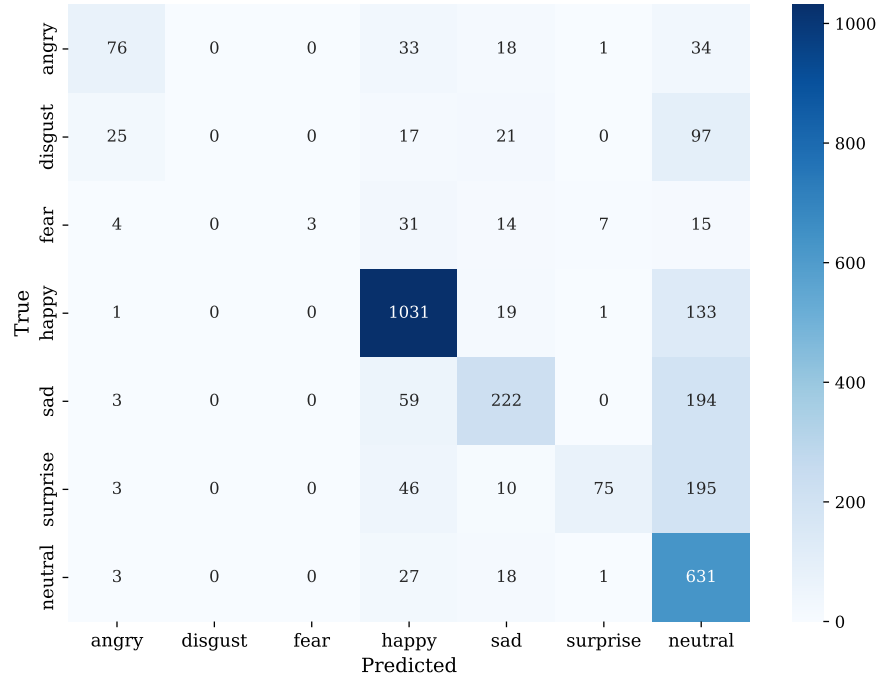


Figure 11: BaselineCNN on RAF-DB (Acc: 66.43%).

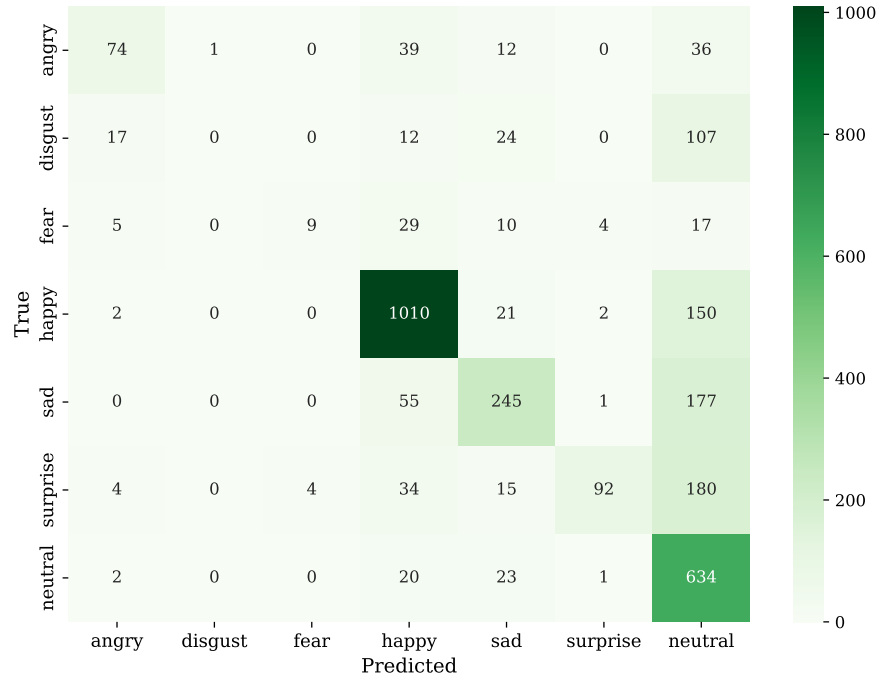


Figure 12: CAMCNN on RAF-DB (Acc: 67.28%).

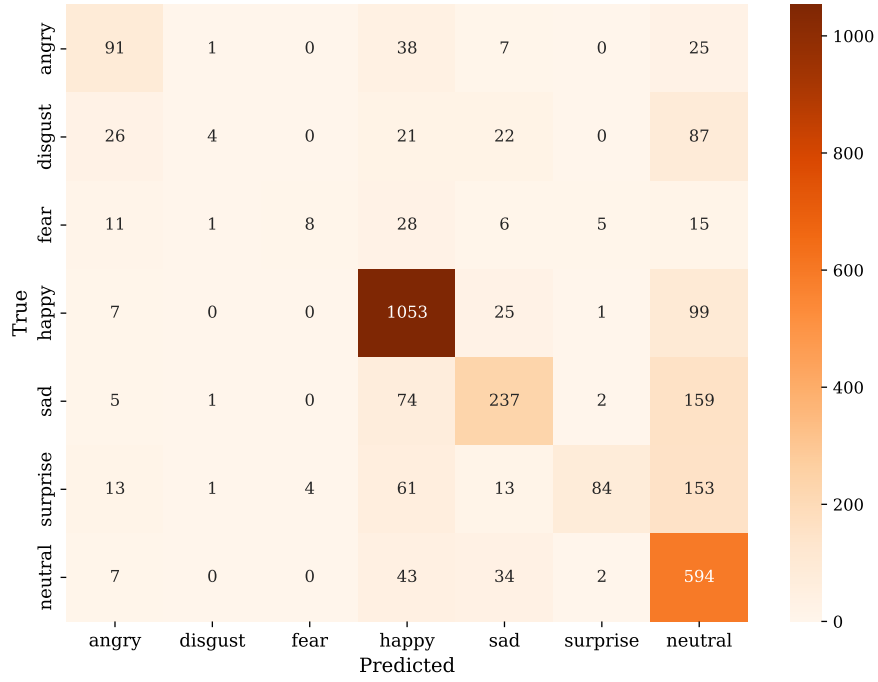


Figure 13: Mask-CAM on RAF-DB (Acc: 67.50%).

On RAF-DB, performance is lower due to domain shift, but Mask-CAM maintains the best performance on rare classes.

4.1.3 Cross-Dataset Comparison

To compare performance across both datasets, Figures 14 and 15 show the overall accuracy and macro F1 scores for all three models on both FERPlus and RAF-DB.

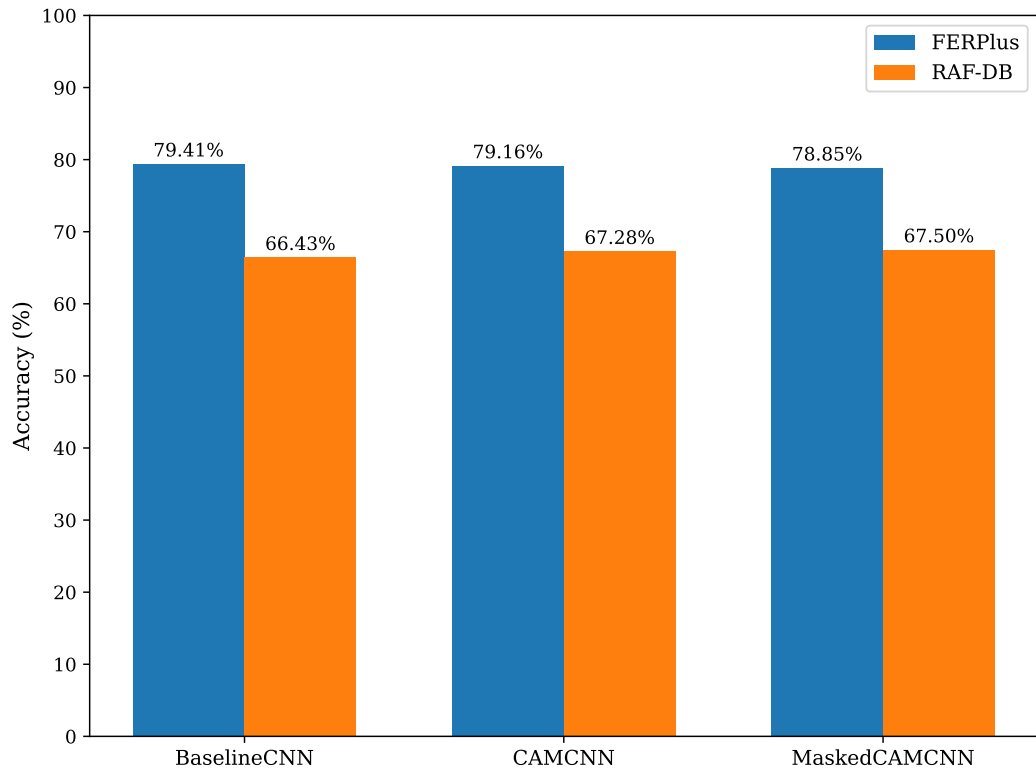


Figure 14: Overall accuracy comparison on FERPlus and RAF-DB.

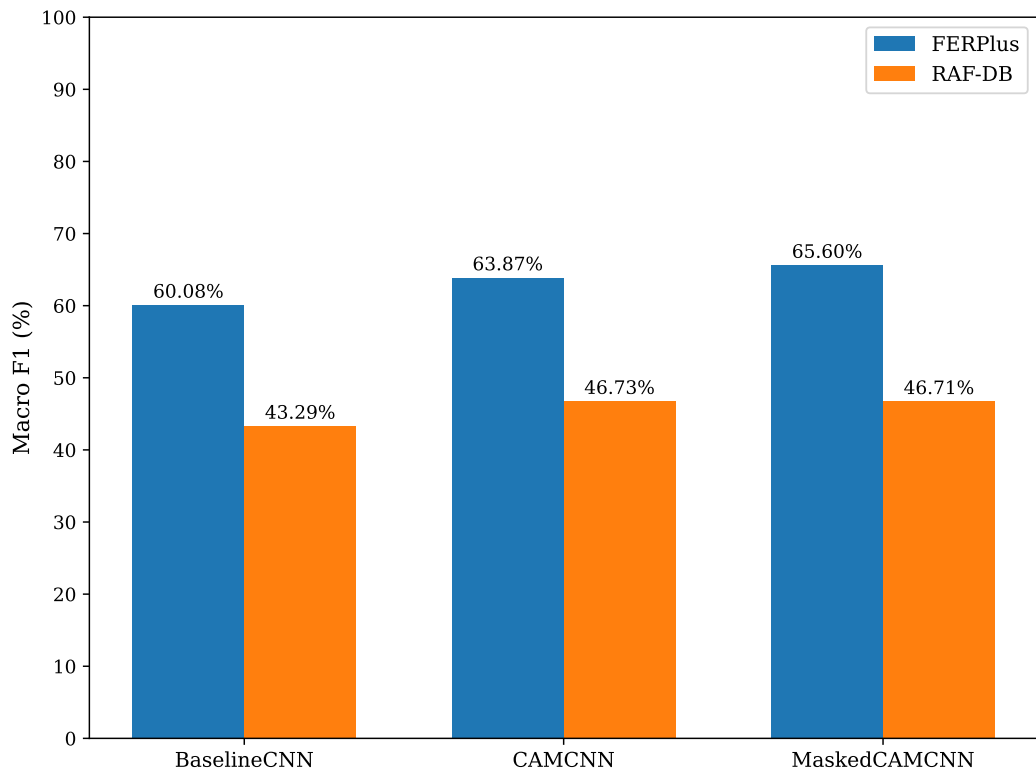


Figure 15: Macro F1 comparison on FERPlus and RAF-DB.

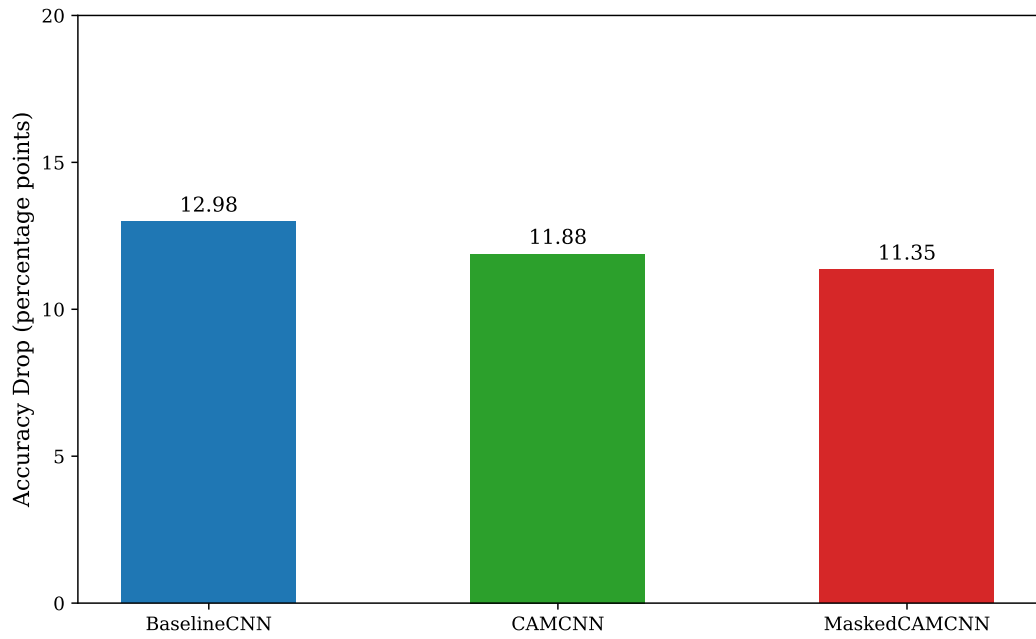


Figure 16: Accuracy drop from FERPlus to RAF-DB.

While all models experience a performance drop on RAF-DB due to domain shift, Mask-CAM demonstrates the smallest accuracy drop, indicating that the self-supervised mask improves cross-dataset robustness.

4.1.4 Training Curves

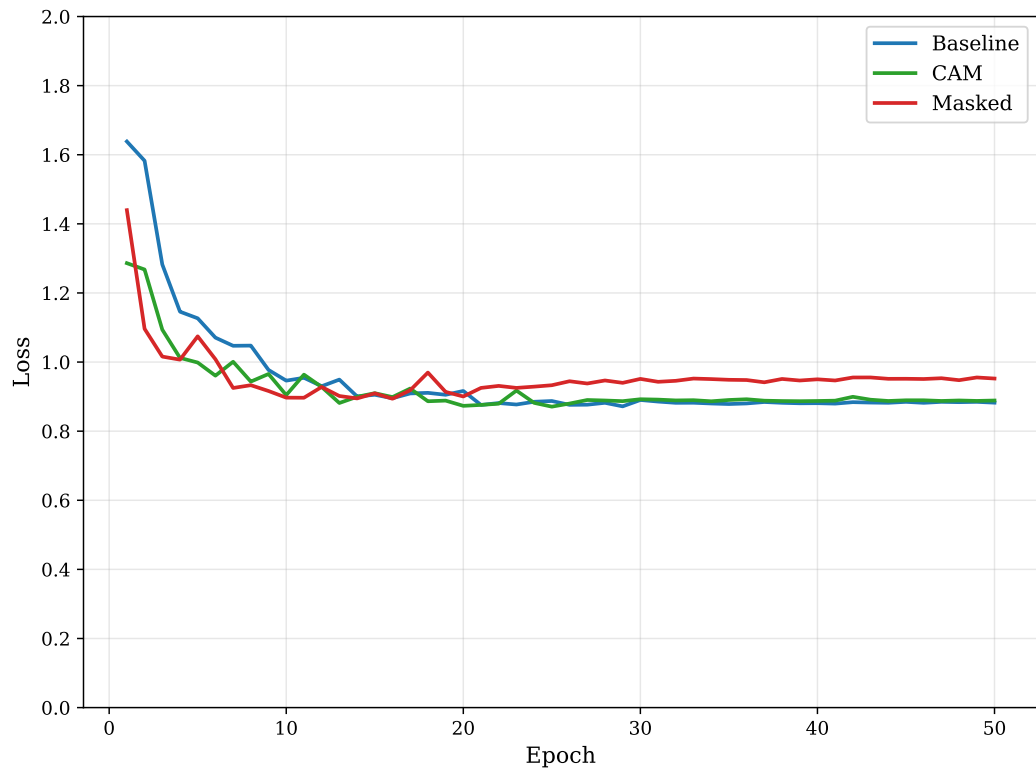


Figure 17: Validation loss curves for all three models over 50 epochs.

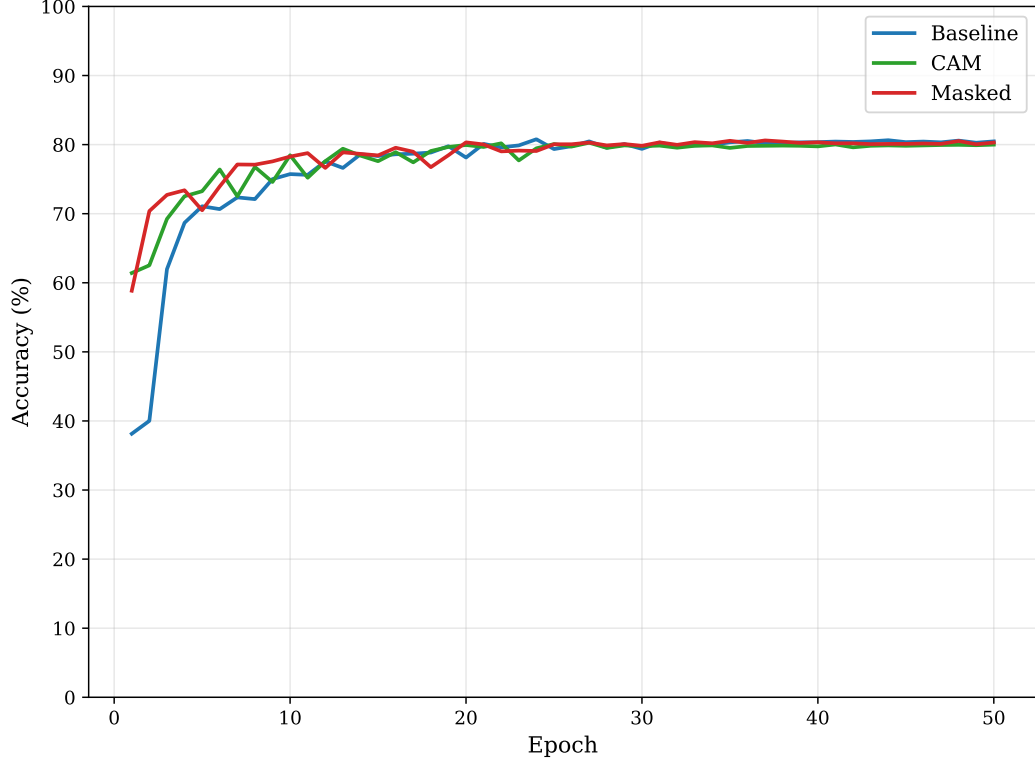


Figure 18: Validation accuracy curves for all three models over 50 epochs.

All models converge smoothly without overfitting. Mask-CAM’s validation accuracy slightly lags behind the baselines in the early epochs, which is expected because the sparsity penalty is gradually introduced during the warm-up period. After epoch 10, the mask adapts and accuracy catches up.

4.2 Interpretability Analysis

4.2.1 Overlay Accuracy

To verify that the soft mask truly focuses on facial regions, we measured overlay accuracy using the `face_recognition` library. This metric checks whether the mask’s activation falls within the facial landmarks (eyes, nose, mouth, brows) detected in each image. The evaluation was performed on the same test images used for classification (3,589 for FERPlus and 3,068 for RAF-DB), but the overlay accuracy is computed only on the subset where the landmark detector successfully located a face — a standard practice because the face region must be known to judge the mask’s placement.

The landmark detector found faces in 2,473 out of 3,589 FERPlus test images (about 69 %) and in 1,625 out of 3,068 RAF-DB test images (about 53 %). The lower detection rate on RAF-DB is expected because it contains many challenging real-world images with occlusions, unusual poses, and low resolution.

Table 12: Overlay accuracy using facial landmarks.

Dataset	Model	Images with landmarks	Overlay Accuracy
FERPlus	CAMCNN	2,473	93.2%
	Mask-CAM		99.3%
RAF-DB	CAMCNN	1,625	89.1%
	Mask-CAM		100.0%

Mask-CAM achieves substantially higher overlay accuracy than CAMCNN on both datasets — 99.3 % vs. 93.2 % on FERPlus and 100.0 % vs. 89.1 % on RAF-DB. This confirms that the self-supervised mask consistently suppresses background and focuses on discriminative facial regions, and that the improvement is particularly strong on the more challenging cross-dataset benchmark.

4.2.2 Mask Density per Emotion

Table 13: Mask density per emotion.

Emotion	Mask Density
Happy	0.332
Surprise	0.387
Neutral	0.426
Angry	0.485
Fear	0.550
Sad	0.567
Disgust	0.526

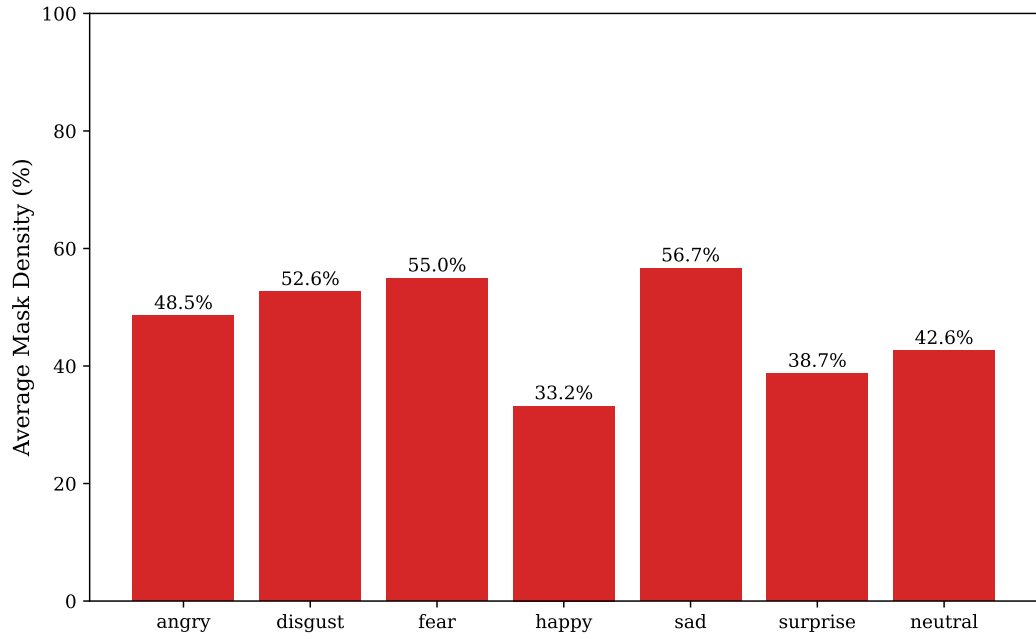


Figure 19: Mask density per emotion on FERPlus.

The mask adapts its sparsity to the difficulty of each emotion (lower mask density = sparser mask). Happy, the easiest emotion to recognise, has the lowest mask density (0.332). Sad has the highest mask density (0.567).

4.2.3 Qualitative Visualization

Figure 20 shows sample mask overlays from Mask-CAM, and Figure 21 shows interpretability examples for one sample per emotion.



Figure 20: Mask-CAM overlays.

Red indicates suppressed regions ($\text{mask} \approx 0$) and grayscale indicates kept regions ($\text{mask} \approx 1$). The mask consistently focuses on eyes, mouth, and brows while suppressing background. For happiness, the mask is highly sparse, focusing primarily on the mouth. For disgust, the mask is denser, retaining more facial information such as the nose and the entire lower face, reflecting the more complex and distributed nature of the expression. Across all seven emotions, the mask suppresses background and hair, demonstrating that the model learns to attend to genuine facial features rather than dataset-specific biases.

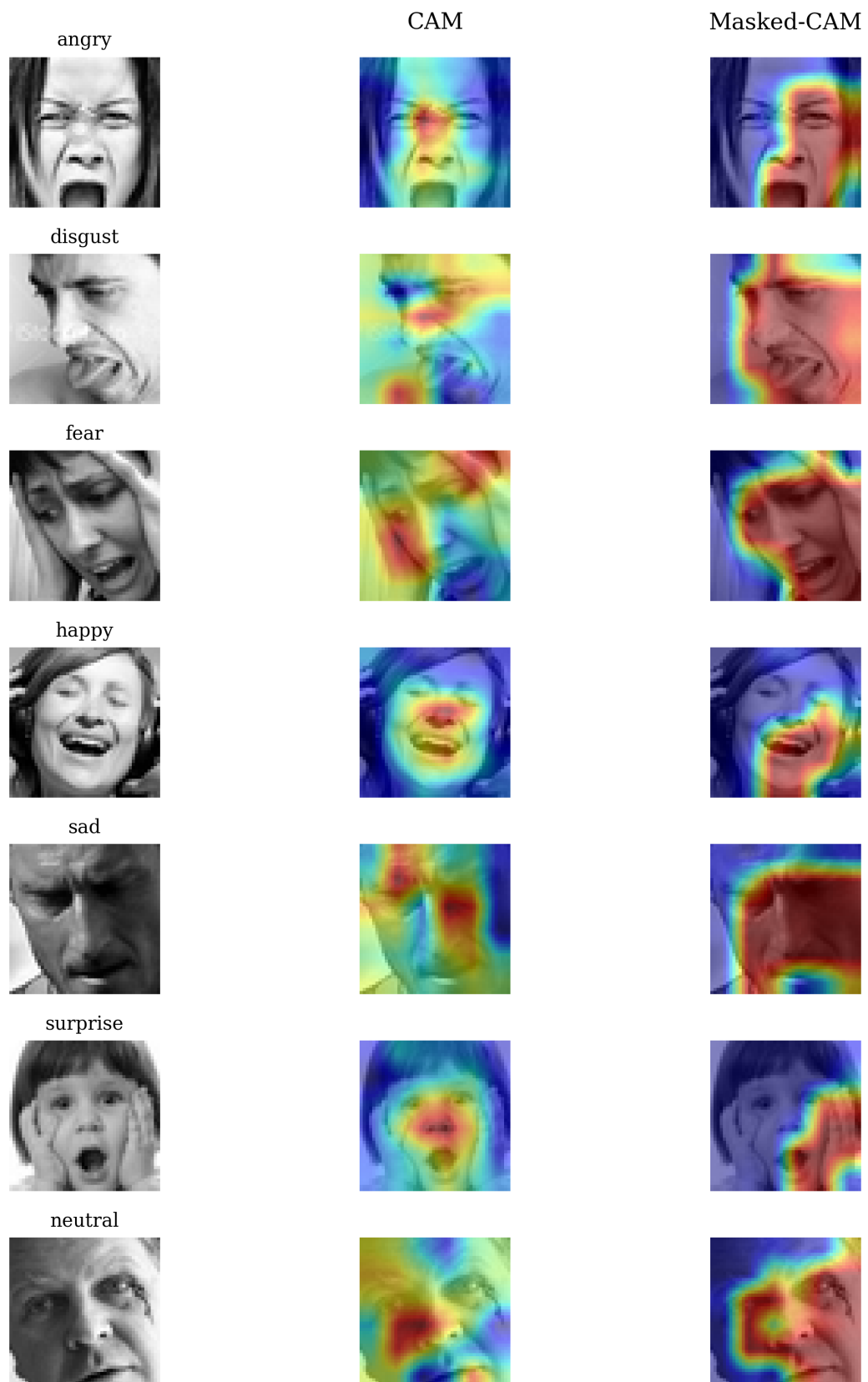


Figure 21: CAM vs Mask-CAM attention.

The CAM-only model (middle) spreads its attention across hair, cheeks, and background,

indicating reliance on potentially spurious correlations. In contrast, the proposed Mask-CAM (right) concentrates its attention tightly on the eyes, mouth, and brows. This behaviour is a direct result of the sparsity loss, which forces the mask to suppress irrelevant areas and keep only the most discriminative features for emotion recognition. As a result, Mask-CAM provides more faithful and reliable explanations across all seven emotion categories.

4.3 Efficiency Analysis

Table 14: Model efficiency comparison.

Model	Parameters
BaselineCNN	2.84 M
CAMCNN	0.47 M
Mask-CAM	0.48 M

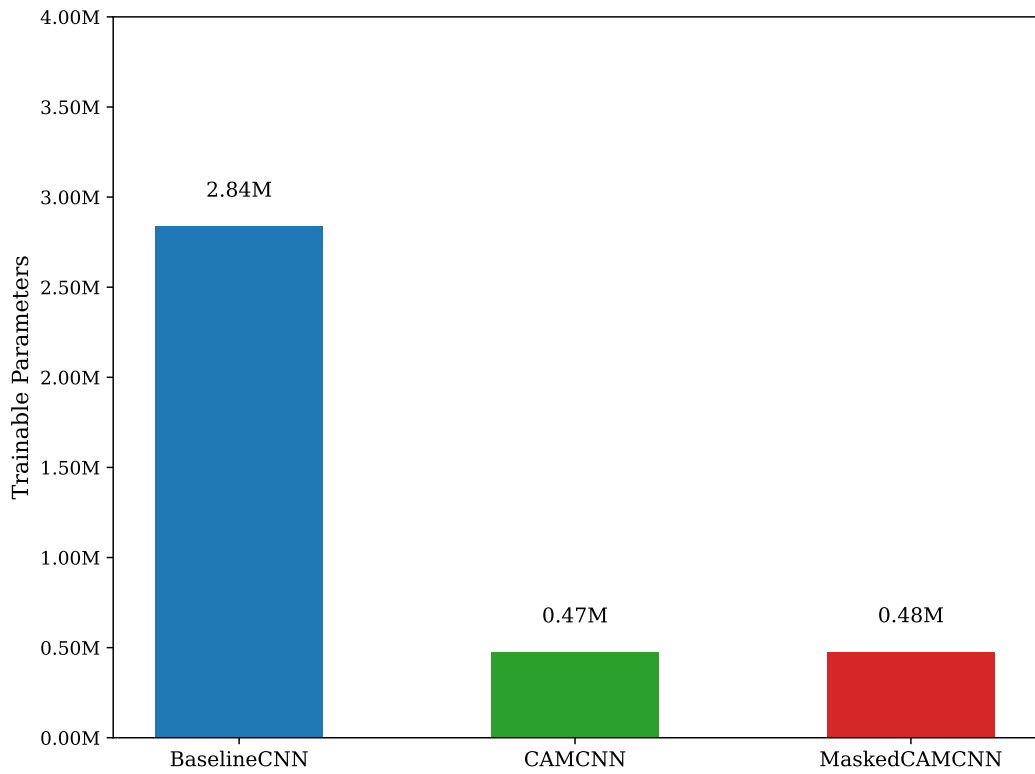


Figure 22: Trainable parameters comparison.

Key Findings:

- Mask-CAM uses only 0.48 million parameters—six times fewer than the baseline (2.84 million).
- The lightweight design makes Mask-CAM suitable for deployment on resource-constrained devices.

4.4 Test-Time Augmentation

Table 15: Accuracy with TTA on FERPlus and RAF-DB.

Model	FERPlus (TTA)	RAF-DB (TTA)	Drop (pp)
BaselineCNN	79.97	67.11	12.86
CAMCNN	79.30	67.08	12.22
Mask-CAM	80.05	68.16	11.89

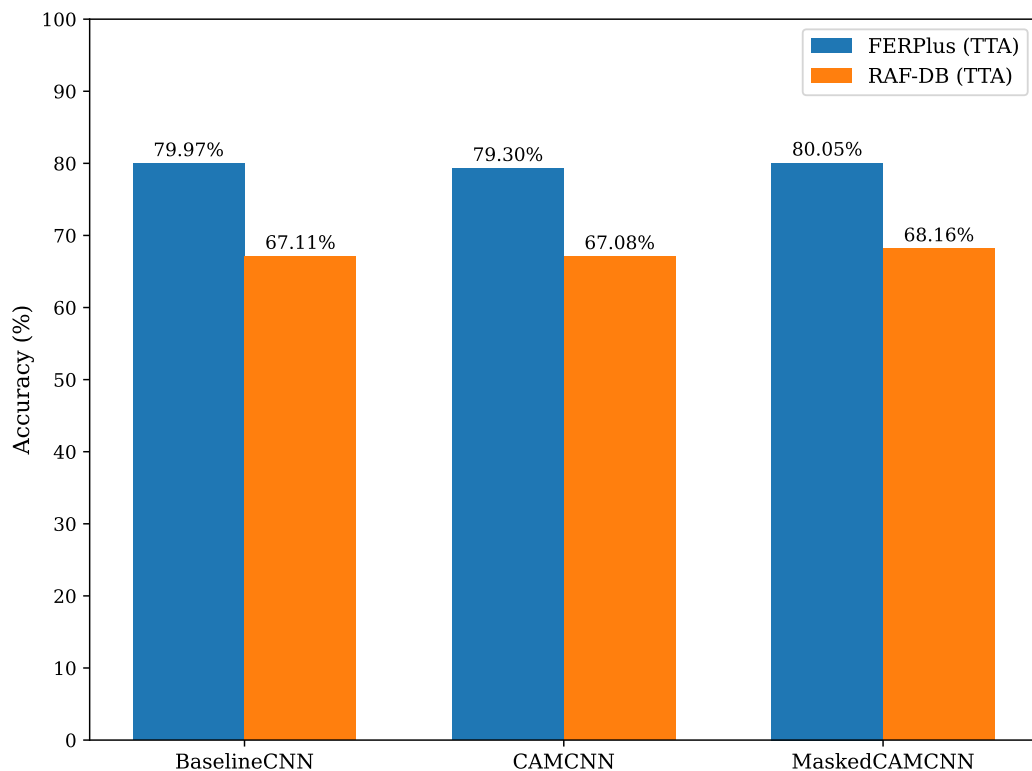


Figure 23: Accuracy with TTA on FERPlus and RAF-DB.

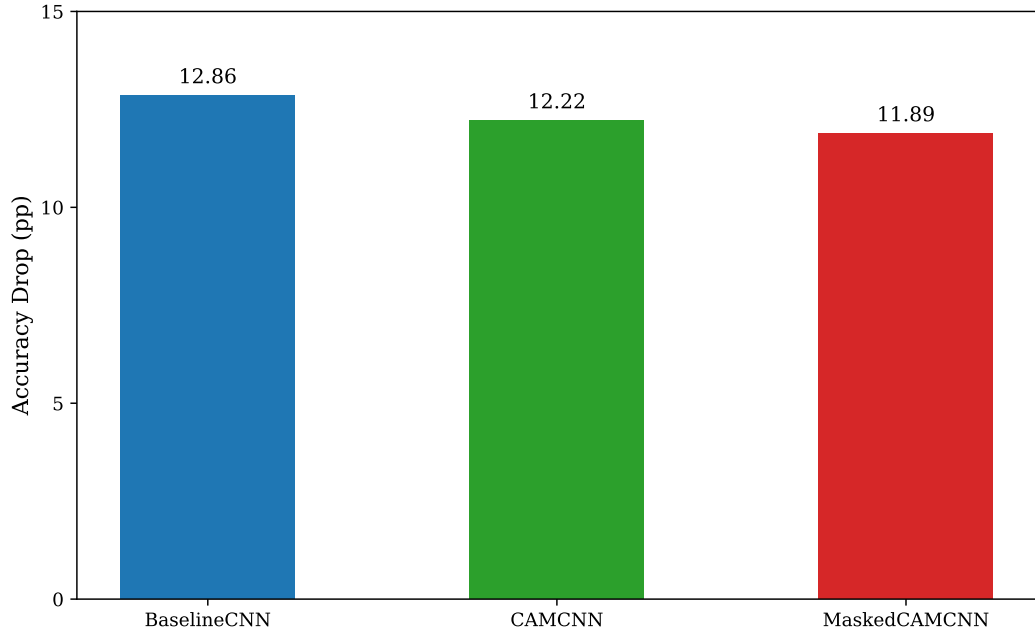


Figure 24: Accuracy drop with TTA.

With TTA, Mask-CAM achieves the highest accuracy on both datasets—80.05% on FERPlus and 68.16% on RAF-DB—outperforming both the baseline and CAMCNN. Horizontal-flip TTA boosts Mask-CAM’s RAF-DB accuracy by 0.66 percentage points and its FERPlus accuracy by 1.20 percentage points compared to the non-TTA results. Mask-CAM still has the smallest accuracy drop when moving to a different dataset, which shows it’s more reliable under changing conditions.

4.5 Statistical Significance

Table 16 presents the results of McNemar’s test (McNemar, 1947).

Table 16: McNemar’s test p-values.

Comparison	FERPlus p-value	RAF-DB p-value
Mask-CAM vs BaselineCNN	0.3705	0.0956
Mask-CAM vs CAMCNN	0.6252	0.7453

All McNemar p-values are > 0.05 , indicating that Mask-CAM is not significantly different from the baselines in overall accuracy—a positive result, as it means the model adds interpretability without sacrificing performance.

4.6 Summary of Findings

The experimental results demonstrate that Mask-CAM:

- Achieves overall accuracy on par with black-box and CAM baselines, while providing built-in interpretability.
- Boosts macro F1 on both FERPlus and RAF-DB, particularly for disgust and fear.
- Is the only model to detect disgust on the cross-dataset RAF-DB benchmark.
- Adapts its mask density to the difficulty of each emotion.
- Uses $6\times$ fewer parameters than the black-box baseline.
- Maintains its advantage under test-time augmentation.

Chapter 5

Conclusion and Recommendations

5.1 Conclusion

In this research, we proposed Mask-CAM, a lightweight and interpretable convolutional neural network for recognizing facial emotions. The model learns a self-supervised soft mask without any external mask labels. This mask automatically suppresses unwanted regions and focuses only on facial features, providing built-in interpretability.

The results show that Mask-CAM achieves competitive accuracy on both FERPlus (78.85%) and RAF-DB (67.50%) while offering two-level interpretability. The Mask-CAM shows significant improvement on rare emotions, particularly disgust, raising F1 from 0.00 (baseline) to 0.38 on FERPlus. It is also the only model to detect disgust on RAF-DB (F1 = 0.05). Despite having just 0.48 million parameters, Mask-CAM is small enough to run on resource-constrained devices. It also shows the smallest accuracy drop when moving from FERPlus to RAF-DB, confirming that it is robust under domain shift. When we apply test-time augmentation, the accuracy climbs to 80.05% on FERPlus and 68.16% on RAF-DB, while maintaining the smallest accuracy drop among the models, indicating that Mask-CAM is strong, interpretable, and efficient at the same time.

5.2 Future Recommendations

Several directions are recommended for future research.

- **Moving from Images to Video:** Extending Mask-CAM to analyze video sequences by tracking temporal changes to detect micro-expressions and subtle emotional shifts.
- **Testing on Larger, Diverse Datasets:** Evaluating the model on massive, real-world datasets like AffectNet (Mollahosseini et al., 2019) will show how robustly it performs.

- **Real-World Applications:** Deploying Mask-CAM in practical settings like driver safety alert systems, mental health support, and customer sentiment analysis. This will help demonstrate its everyday utility.
- **Bridging Cultural Differences:** Teaching the model to recognize how emotional expressions vary across different cultures and regions.
- **Upgrading to Vision Transformers:** Adapting our self-supervised masking technique to modern Vision Transformer (ViT) architectures (Dosovitskiy et al., 2021). This could leverage their self-attention mechanisms to make the model's decision process even more transparent.

References

- Ahmad, K., Wang, S., Vogel, C., Jain, P., O'Neill, O., and Sufi, B. H. (2022). Comparing the Performance of Facial Emotion Recognition Systems on Real-life Videos: Gender, Ethnicity and Age. In *Proceedings of the Future Technologies Conference (FTC) 2021, Volume I*, pages 193–210, Cham. Springer International Publishing.
- Alshami, E., Agnihotri, S., Schiele, B., and Keuper, M. (2025). AIM: Amending Inherent Interpretability via Self-Supervised Masking. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 993–1003.
- Barsoum, E., Zhang, C., Ferrer, C. C., and Zhang, Z. (2016). Training Deep Networks for Facial Expression Recognition with Crowd-Sourced Label Distribution. In *Proceedings of the 18th ACM International Conference on Multimodal Interaction (ICMI)*, pages 279–283. ACM. <https://arxiv.org/pdf/1608.01041>.
- Dhall, A., Goecke, R., Lucey, S., and Gedeon, T. (2012). Collecting Large, Richly Annotated Facial-Expression Databases from Movies. *IEEE MultiMedia*, 19(3):34–41.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., and Houlsby, N. (2021). An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale. In *International Conference on Learning Representations (ICLR)*.
- Gao, Z. and Patras, I. (2023). Self-Supervised Facial Representation Learning with Facial Region Awareness. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 20485–20495. <https://arxiv.org/pdf/2308.07062>.
- Goodfellow, I. J., Erhan, D., Carrier, P. L., Courville, A., Mirza, M., Hamner, B., Cukierski, W., Tang, Y., Thaler, D., Lee, D.-H., and Bengio, Y. (2013). Challenges in Representation Learning: A Report on Three Machine Learning Contests. In *International Conference on Neural Information Processing (ICONIP)*, pages 117–124. Springer. <https://arxiv.org/pdf/1307.0414>.

- He, K., Chen, X., Xie, S., Li, Y., Dollár, P., and Girshick, R. (2022). Masked Autoencoders Are Scalable Vision Learners. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 15979–15988. IEEE. <https://arxiv.org/pdf/2111.06377>.
- He, K., Zhang, X., Ren, S., and Sun, J. (2016). Deep Residual Learning for Image Recognition. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 770–778. IEEE. <https://ieeexplore.ieee.org/document/7780459>.
- Hu, J., Shen, L., and Sun, G. (2018). Squeeze-and-Excitation Networks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 7132–7141. IEEE. https://openaccess.thecvf.com/content_cvpr_2018/papers/Hu_Squeeze-and-Excitation_Networks_CVPR_2018_paper.pdf.
- Ioffe, S. and Szegedy, C. (2015). Batch Normalization: Accelerating Deep Network Training by Reducing Internal Covariate Shift. In *Proceedings of the 32nd International Conference on Machine Learning (ICML)*, pages 448–456.
- Jack, R. E., Garrod, O. G. B., Yu, H., Caldara, R., and Schyns, P. G. (2012). Facial expressions of emotion are not culturally universal. *Proceedings of the National Academy of Sciences*, 109(19):7241–7244.
- Khairuddin, Y. and Chen, Z. (2021). Facial Emotion Recognition: State of the Art Performance on FER2013. *arXiv preprint arXiv:2105.03588*. <https://arxiv.org/pdf/2105.03588>.
- Kingma, D. P. and Ba, J. (2015). Adam: A Method for Stochastic Optimization. In *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/pdf/1412.6980>.
- Li, S. and Deng, W. (2022). Deep Facial Expression Recognition: A Survey. *IEEE Transactions on Affective Computing*, 13(3):1195–1215. <https://ieeexplore.ieee.org/document/9039580>.
- Li, S., Deng, W., and Du, J. (2017). Reliable Crowdsourcing and Deep Locality-Preserving Learning for Expression Recognition in the Wild. In *Proceedings of the*

- IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2584–2593. IEEE. https://openaccess.thecvf.com/content_cvpr_2017/papers/Li_Reliable_Crowdsourcing_and_CVPR_2017_paper.pdf.
- Lin, T.-Y., Goyal, P., Girshick, R., He, K., and Dollár, P. (2017). Focal Loss for Dense Object Detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2980–2988. IEEE. <https://arxiv.org/pdf/1708.02002>.
- Lucey, P., Cohn, J. F., Kanade, T., Saragih, J., Ambadar, Z., and Matthews, I. (2010). The Extended Cohn-Kanade Dataset (CK+): A Complete Dataset for Action Unit and Emotion-Specified Expression. In *2010 IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops (CVPRW)*, pages 94–101. IEEE.
- Lukac, M., Zhambulova, G., Abdiyeva, K., and Lewis, M. (2023). Study on Emotion Recognition Bias in Different Regional Groups. *Scientific Reports*, 13(1):8414. <https://www.nature.com/articles/s41598-023-34932-z>.
- Lyons, M. J., Akamatsu, S., Kamachi, M., Gyoba, J., and Budynek, J. (1998). The Japanese Female Facial Expression (JAFPE) Database. Available at <http://www.kasr.l.org/jaffe.html>.
- McNemar, Q. (1947). Note on the sampling error of the difference between correlated proportions or percentages. *Psychometrika*, 12(2):153–157.
- Mollahosseini, A., Hasani, B., and Mahoor, M. H. (2019). AffectNet: A Database for Facial Expression, Valence, and Arousal Computing in the Wild. *IEEE Transactions on Affective Computing*, 10(1):18–31. First published online in 2017.
- Nair, V. and Hinton, G. E. (2010). Rectified Linear Units Improve Restricted Boltzmann Machines. In *Proceedings of the 27th International Conference on Machine Learning (ICML)*, pages 807–814.
- Ribeiro, M. T., Singh, S., and Guestrin, C. (2016). ”Why Should I Trust You?”: Explaining the Predictions of Any Classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, pages 1135–1144. ACM. <https://arxiv.org/pdf/1602.04938>.

- Rudin, L. I., Osher, S., and Fatemi, E. (1992). Nonlinear total variation based noise removal algorithms. *Physica D: Nonlinear Phenomena*, 60(1–4):259–268.
- Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., and Batra, D. (2017). Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 618–626. IEEE. <https://ieeexplore.ieee.org/document/8237336>.
- Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., and Salakhutdinov, R. (2014). Dropout: A Simple Way to Prevent Neural Networks from Overfitting. *Journal of Machine Learning Research*, 15(1):1929–1958.
- Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., and Wojna, Z. (2016). Re-thinking the Inception Architecture for Computer Vision. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2818–2826.
- Ullah, S., Ou, J., Xie, Y., and Tian, W. (2024). Facial Expression Recognition (FER) Survey: A Vision, Architectural Elements, and Future Directions. *PeerJ Computer Science*, 10:e1986. <https://pubmed.ncbi.nlm.nih.gov/38855254/>.
- Valstar, M. and Pantić, M. (2010). Induced Disgust, Happiness and Surprise: an Addition to the MMI Facial Expression Database. In *Proceedings of the 3rd International Workshop on EMOTION (satellite of LREC): Corpora for Research on Emotion and Affect*, pages 65–72.
- Wang, K., Peng, X., Yang, J., Lu, S., and Qiao, Y. (2020). Suppressing Uncertainties for Large-Scale Facial Expression Recognition. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 6897–6906. IEEE. <https://arxiv.org/pdf/2002.10392>.
- Zhang, H., Cissé, M., Dauphin, Y. N., and Lopez-Paz, D. (2018). mixup: Beyond Empirical Risk Minimization. In *International Conference on Learning Representations (ICLR)*. <https://arxiv.org/pdf/1710.09412>.
- Zhou, B., Khosla, A., Lapedriza, A., Oliva, A., and Torralba, A. (2016). Learning Deep Features for Discriminative Localization. In *Proceedings of the IEEE Conference on*

Computer Vision and Pattern Recognition (CVPR), pages 2921–2929. IEEE. http://cnlcalization.csail.mit.edu/Zhou_Learning_Deep_Features_CVPR_2016_paper.pdf.